Main Category	Category	Terms
Disease	OMIM_DISEASE	Oocyte/zygote/embryo maturation arrest 21
Disease	UP_KW_DISEASE	Disease variant
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Cell cycle
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA damage
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA repair
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Chromosome
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoskeleton
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Kinase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Serine/threonine-protein kinase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Transferase
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_KW_PTM	Isopeptide bond
Functional Annotations	UP_SEQ_FEATURE	ACT_SITE:Proton acceptor
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Polar residues
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cter in ubiquitin)
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Protein kinase
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:D->A: Abolishes kinase activity.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:F->A: Impairs nuclear export.
Functional	UP_SEQ_FEATURE	MUTAGEN:K->R: Abolishes kinase activity.

Main Category	Category	Terms
Annotations		
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Enhances stability of the protein, probably by preventing ubiquitination at this site.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Strong reduction of chromatin-associated CHK1 ubiquitination.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:M->A: Impairs nuclear export.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:R->E: In 3RE mutant. Disrupts the folding and/or conformation, allowing increased accessibility to FBXO6 component of SCF-type E3 ubiquitin ligase complex; when associated with E-372 and E-376.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:R->E: In 3RE mutant. Disrupts the folding and/or conformation, allowing increased accessibility to FBXO6 component of SCF-type E3 ubiquitin ligase complex; when associated with E-372 and E-379.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:R->E: In 3RE mutant. Disrupts the folding and/or conformation, allowing increased accessibility to FBXO6 component of SCF-type E3 ubiquitin ligase complex; when associated with E-376 and E-379.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: Abrogates interaction with RAD51; when associated with A-317. Reduces phosphorylation and impairs activation by hydroxyurea and ionizing radiation. Impairs interaction with YWHAZ which is required for nuclear retention after checkpoint activation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: Abrogates interaction with RAD51; when associated with A-345. Reduces phosphorylation and impairs activation by hydroxyurea and ionizing radiation. Abrogates nuclear retention upon checkpoint activation. Impairs interaction with FBXO6.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: No effect on phosphorylation induced by hydroxyurea.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->E: Enhances interaction with RAD51; when associated with E-317.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->E: Enhances interaction with RAD51; when associated with E-345.
Functional Annotations	UP_SEQ_FEATURE	REGION:Autoinhibitory region
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with CLSPN
Gene Ontology	GOTERM_BP_DIRECT	DNA damage checkpoint signaling
Gene Ontology	GOTERM_BP_DIRECT	G2/M transition of mitotic cell cycle

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	inner cell mass cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	DNA replication
Gene Ontology	GOTERM_BP_DIRECT	DNA repair
Gene Ontology	GOTERM_BP_DIRECT	chromatin remodeling
Gene Ontology	GOTERM_BP_DIRECT	protein phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	DNA damage response
Gene Ontology	GOTERM_BP_DIRECT	nucleus organization
Gene Ontology	GOTERM_BP_DIRECT	nuclear envelope organization
Gene Ontology	GOTERM_BP_DIRECT	mitotic nuclear membrane disassembly
Gene Ontology	GOTERM_BP_DIRECT	mitotic G2 DNA damage checkpoint signaling
Gene Ontology	GOTERM_BP_DIRECT	regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	regulation of double-strand break repair via homologous recombination
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of G2/M transition of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	peptidyl-threonine phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	regulation of cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	signal transduction in response to DNA damage
Gene Ontology	GOTERM_BP_DIRECT	mitotic G2/M transition checkpoint
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cell cycle
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of gene expression, epigenetic
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of mitotic nuclear division

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	regulation of mitotic centrosome separation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of G0 to G1 transition
Gene Ontology	GOTERM_BP_DIRECT	cellular response to mechanical stimulus
Gene Ontology	GOTERM_BP_DIRECT	replicative senescence
Gene Ontology	GOTERM_BP_DIRECT	regulation of signal transduction by p53 class mediator
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell cycle phase transition
Gene Ontology	GOTERM_BP_DIRECT	apoptotic process involved in development
Gene Ontology	GOTERM_CC_DIRECT	chromosome, telomeric region
Gene Ontology	GOTERM_CC_DIRECT	chromatin
Gene Ontology	GOTERM_CC_DIRECT	condensed nuclear chromosome
Gene Ontology	GOTERM_CC_DIRECT	extracellular space
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	replication fork
Gene Ontology	GOTERM_CC_DIRECT	chromosome
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	centrosome
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	cytoskeleton
Gene Ontology	GOTERM_CC_DIRECT	protein-containing complex
Gene Ontology	GOTERM_CC_DIRECT	intracellular membrane-bounded organelle

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	nucleotide binding
Gene Ontology	GOTERM_MF_DIRECT	protein kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein serine/threonine kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	ATP binding
Gene Ontology	GOTERM_MF_DIRECT	kinase activity
Gene Ontology	GOTERM_MF_DIRECT	transferase activity
Gene Ontology	GOTERM_MF_DIRECT	protein domain specific binding
Gene Ontology	GOTERM_MF_DIRECT	histone H3T11 kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein serine kinase activity
General Annotations	CHROMOSOME	11
General Annotations	CYTOGENETIC_LOCATION	11q24.2
General Annotations	ENTREZ_GENE_SUMMARY	The protein encoded by this gene belongs to the Ser/Thr protein kinase family. It is required for checkpoint mediated cell cycle arrest in response to DNA damage or the presence of unreplicated DNA. This protein acts to integrate signals from ATM and ATR, two cell cycle proteins involved in DNA damage responses, that also associate with chromatin in meiotic prophase I. Phosphorylation of CDC25A protein phosphatase by this protein is required for cells to delay cell cycle progression in response to double-strand DNA breaks. Several alternatively spliced transcript variants have been found for this gene. [provided by RefSeq, Oct 2011]
General Annotations	UP_COMMENT_ACTIVITY_REGULATION	Activated through phosphorylation predominantly by ATR but also by ATM in response to DNA damage or inhibition of DNA replication (PubMed:11390642, PubMed:12588868, PubMed:12676583, PubMed:12676962, PubMed:15665856, PubMed:19716789). Activation is modulated by several mediators including CLSPN, BRCA1 and FEM1B (PubMed:11836499, PubMed:12766152, PubMed:16963448, PubMed:19330022). Proteolytic cleavage at the C-terminus by SPRTN during normal DNA replication activates the protein kinase activity (PubMed:31316063). {ECO:0000269|

Main Category	Category	Terms
		PubMed:11390642, ECO:0000269|PubMed:11836499, ECO:0000269|PubMed:12588868, ECO:0000269|PubMed:12676583, ECO:0000269|PubMed:12676962, ECO:0000269|PubMed:12766152, ECO:0000269|PubMed:15665856, ECO:0000269|PubMed:16963448, ECO:0000269|PubMed:19330022, ECO:0000269|PubMed:19716789, ECO:0000269|PubMed:31316063}.
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing; Named isoforms=3; Name=1; IsoId=O14757-1; Sequence=Displayed; Name=2; Synonyms=Chk1-short {ECO:0000303|PubMed:22184239}, Chk1-S {ECO:0000303|PubMed:22184239}; IsoId=O14757-2; Sequence=VSP_044008, VSP_044009; Name=3; IsoId=O14757-3; Sequence=VSP_045075;
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-seryl-[protein] + ATP = O-phospho-L-seryl-[protein] + ADP + H(+); Xref=Rhea:RHEA:17989, Rhea:RHEA-COMP:9863, Rhea:RHEA- COMP:11604, ChEBI:CHEBI:15378, ChEBI:CHEBI:29999, ChEBI:CHEBI:30616, ChEBI:CHEBI:83421, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000256|ARBA:ARBA00048679};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-seryl-[protein] + ATP = O-phospho-L-seryl-[protein] + ADP + H(+); Xref=Rhea:RHEA:17989, Rhea:RHEA-COMP:9863, Rhea:RHEA- COMP:11604, ChEBI:CHEBI:15378, ChEBI:CHEBI:29999, ChEBI:CHEBI:30616, ChEBI:CHEBI:83421, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000269|PubMed:10673501, ECO:0000269|PubMed:12660173, ECO:0000269|PubMed:14559997, ECO:0000269|PubMed:15659650, ECO:0000269|PubMed:15665856, ECO:0000269|PubMed:16511572, ECO:0000269|PubMed:16963448, ECO:0000269|PubMed:17296736, ECO:0000269|PubMed:17380128, ECO:0000269|PubMed:18317453, ECO:0000269|PubMed:18728393, ECO:0000269|PubMed:31316063, ECO:0000269|PubMed:9278511};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-threonyl-[protein] + ATP = O-phospho-L-threonyl-[protein] + ADP + H(+); Xref=Rhea:RHEA:46608, Rhea:RHEA-COMP:11060, Rhea:RHEA- COMP:11605, ChEBI:CHEBI:15378, ChEBI:CHEBI:30013, ChEBI:CHEBI:30616, ChEBI:CHEBI:61977, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000256|ARBA:ARBA00047899};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-threonyl-[protein] + ATP = O-phospho-L-threonyl-[protein] + ADP + H(+); Xref=Rhea:RHEA:46608, Rhea:RHEA-COMP:11060, Rhea:RHEA- COMP:11605, ChEBI:CHEBI:15378, ChEBI:CHEBI:30013, ChEBI:CHEBI:30616, ChEBI:CHEBI:61977, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000269|PubMed:10673501, ECO:0000269|PubMed:12660173, ECO:0000269|PubMed:14559997, ECO:0000269|PubMed:15659650, ECO:0000269|PubMed:15665856, ECO:0000269|PubMed:16511572, ECO:0000269|PubMed:16963448, ECO:0000269|PubMed:17296736, ECO:0000269|PubMed:17380128, ECO:0000269|PubMed:18317453,

Main Category	Category	Terms
		ECO:0000269|PubMed:18728393, ECO:0000269|PubMed:31316063, ECO:0000269|PubMed:9278511};
General Annotations	UP_COMMENT_DISEASE	Oocyte/zygote/embryo maturation arrest 21 (OZEMA21) [MIM:620610]: An autosomal dominant, female infertility disorder characterized by zygote development arrest due to failure of pronuclei fusion. {ECO:0000269|PubMed:33948904, ECO:0000269|PubMed:33953335}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DOMAIN	The autoinhibitory region (AIR) inhibits the activity of the kinase domain. {ECO:0000269|PubMed:14681223}.
General Annotations	UP_COMMENT_FUNCTION	Serine/threonine-protein kinase which is required for checkpoint-mediated cell cycle arrest and activation of DNA repair in response to the presence of DNA damage or unreplicated DNA (PubMed:11535615, PubMed:12399544, PubMed:12446774, PubMed:14559997, PubMed:14988723, PubMed:15311285, PubMed:15650047, PubMed:15665856, PubMed:32357935). May also negatively regulate cell cycle progression during unperturbed cell cycles (PubMed:11535615, PubMed:12399544, PubMed:12446774, PubMed:14559997, PubMed:14988723, PubMed:15311285, PubMed:15650047, PubMed:15665856). This regulation is achieved by a number of mechanisms that together help to preserve the integrity of the genome (PubMed:11535615, PubMed:12399544, PubMed:12446774, PubMed:14559997, PubMed:14988723, PubMed:15311285, PubMed:15650047, PubMed:15665856). Recognizes the substrate consensus sequence [R-X-X- S/T] (PubMed:11535615, PubMed:12399544, PubMed:12446774, PubMed:14559997, PubMed:14988723, PubMed:15311285, PubMed:15650047, PubMed:15665856). Binds to and phosphorylates CDC25A, CDC25B and CDC25C (PubMed:12676583, PubMed:12676925, PubMed:12759351, PubMed:14559997, PubMed:14681206, PubMed:19734889, PubMed:9278511). Phosphorylation of CDC25A at 'Ser-178' and 'Thr-507' and phosphorylation of CDC25C at 'Ser-216' creates binding sites for 14-3-3 proteins which inhibit CDC25A and CDC25C (PubMed:9278511). Phosphorylation of CDC25A at 'Ser- 76', 'Ser-124', 'Ser-178', 'Ser-279' and 'Ser-293' promotes proteolysis of CDC25A (PubMed:12676583, PubMed:12676925, PubMed:12759351, PubMed:14681206, PubMed:19734889, PubMed:9278511). Phosphorylation of CDC25A at 'Ser-76' primes the protein for subsequent phosphorylation at 'Ser-79', 'Ser-82' and 'Ser-88' by NEK11, which is required for polyubiquitination and degradation of CDCD25A (PubMed:19734889, PubMed:20090422, PubMed:9278511). Inhibition of CDC25 leads to increased inhibitory tyrosine phosphorylation of CDK-cyclin complexes and blocks cell cycle progression (PubMed:9278511). Also phosphorylates NEK6

Main Category	Category	Terms
		(PubMed:18728393). Binds to and phosphorylates RAD51 at 'Thr-309', which promotes the release of RAD51 from BRCA2 and enhances the association of RAD51 with chromatin, thereby promoting DNA repair by homologous recombination (PubMed:15665856). Phosphorylates multiple sites within the C-terminus of TP53, which promotes activation of TP53 by acetylation and promotes cell cycle arrest and suppression of cellular proliferation (PubMed:10673501, PubMed:15659650, PubMed:16511572). Also promotes repair of DNA cross-links through phosphorylation of FANCE (PubMed:17296736). Binds to and phosphorylates TLK1 at 'Ser-743', which prevents the TLK1-dependent phosphorylation of the chromatin assembly factor ASF1A (PubMed:12660173, PubMed:12955071). This may enhance chromatin assembly both in the presence or absence of DNA damage (PubMed:12660173, PubMed:12955071). May also play a role in replication fork maintenance through regulation of PCNA (PubMed:18451105). May regulate the transcription of genes that regulate cell-cycle progression through the phosphorylation of histones (By similarity). Phosphorylates histone H3.1 (to form H3T11ph), which leads to epigenetic inhibition of a subset of genes (By similarity). May also phosphorylate RB1 to promote its interaction with the E2F family of transcription factors and subsequent cell cycle arrest (PubMed:17380128). Phosphorylates SPRTN, promoting SPRTN recruitment to chromatin (PubMed:31316063). Reduces replication stress and activates the G2/M checkpoint, by phosphorylating and inactivating PABIR1/FAM122A and promoting the serine/threonine-protein phosphatase 2A-mediated dephosphorylation and stabilization of WEE1 levels and activity (PubMed:33108758). {ECO:0000250|UniProtKB:O35280, ECO:0000269|PubMed:10673501, ECO:0000269|PubMed:11535615, ECO:0000269|PubMed:12399544, ECO:0000269|PubMed:12446774, ECO:0000269|PubMed:12660173, ECO:0000269|PubMed:12676583, ECO:0000269|PubMed:12676925, ECO:0000269|PubMed:12759351, ECO:0000269|PubMed:12955071, ECO:0000269|PubMed:14559997, ECO:0000269|PubMed:14681206, ECO:0000269|PubMed:14988723, ECO:0000269|PubMed:15311285, ECO:0000269|PubMed:15650047, ECO:0000269|PubMed:15659650, ECO:0000269|PubMed:15665856, ECO:0000269|PubMed:16511572, ECO:0000269|PubMed:17296736, ECO:0000269|PubMed:17380128, ECO:0000269|PubMed:18451105, ECO:0000269|PubMed:18728393, ECO:0000269|PubMed:19734889, ECO:0000269|PubMed:20090422, ECO:0000269|PubMed:31316063, ECO:0000269|PubMed:32357935, ECO:0000269|PubMed:33108758, ECO:0000269|PubMed:9278511}.
General Annotations	UP_COMMENT_FUNCTION	[Isoform 2]: Endogenous repressor of isoform 1, interacts with, and antagonizes CHK1 to promote the S to G2/M phase transition. {ECO:0000269|PubMed:22184239}.
General	UP_COMMENT_INTERACTION	O14757; P38398: BRCA1; NbExp=3; IntAct=EBI-974488,

Main Category	Category	Terms
Annotations		EBI-349905; O14757; P30307: CDC25C; NbExp=2; IntAct=EBI-974488, EBI-974439; O14757; Q9HAW4: CLSPN; NbExp=5; IntAct=EBI-974488, EBI-1369377; O14757; Q9UJM3: ERRFI1; NbExp=2; IntAct=EBI-974488, EBI-2941912; O14757; Q30154: HLA-DRB5; NbExp=2; IntAct=EBI-974488, EBI-2798348; O14757; P08238: HSP90AB1; NbExp=3; IntAct=EBI-974488, EBI-352572; O14757; Q06609: RAD51; NbExp=3; IntAct=EBI-974488, EBI-297202; O14757; P06400: RB1; NbExp=3; IntAct=EBI-974488, EBI-491274; O14757; Q9HCE7-2: SMURF1; NbExp=4; IntAct=EBI-974488, EBI-9845742; O14757; Q9UNS1: TIMELESS; NbExp=2; IntAct=EBI-974488, EBI-2212315; O14757; P31946: YWHAB; NbExp=2; IntAct=EBI-974488, EBI-359815; O14757; P61981: YWHAG; NbExp=8; IntAct=EBI-974488, EBI-359832;
General Annotations	UP_COMMENT_PTM	Phosphorylated by ATR in a RAD17-dependent manner in response to ultraviolet irradiation and inhibition of DNA replication (PubMed:10859164, PubMed:11390642, PubMed:12446774, PubMed:12588868, PubMed:12676583, PubMed:12676925, PubMed:12676962, PubMed:14681223, PubMed:14988723, PubMed:15650047, PubMed:15707391, PubMed:15870257, PubMed:25083873, PubMed:31316063). Phosphorylated by ATM in response to ionizing irradiation (PubMed:12588868, PubMed:12676583). ATM and ATR can both phosphorylate Ser-317 and Ser-345 and this results in enhanced kinase activity (PubMed:11390642, PubMed:12446774, PubMed:12588868, PubMed:12660173, PubMed:12676583, PubMed:12676962, PubMed:14657349, PubMed:15665856, PubMed:15707391, PubMed:15870257, PubMed:25083873). Phosphorylation at Ser-345 induces a change in the conformation of the protein, activates the kinase activity and is a prerequisite for interaction with FBXO6 and subsequent ubiquitination at Lys-436 (PubMed:19716789). Phosphorylation at Ser-345 also increases binding to 14-3-3 proteins and promotes nuclear retention (PubMed:12676962). Conversely, dephosphorylation at Ser-345 by PPM1D may contribute to exit from checkpoint mediated cell cycle arrest (PubMed:15870257). Phosphorylation at Ser-280 by AKT1/PKB, may promote mono and/or diubiquitination. Also phosphorylated at undefined residues during mitotic arrest, resulting in decreased activity (By similarity). {ECO:0000250|UniProtKB:O35280, ECO:0000269|PubMed:10859164, ECO:0000269|PubMed:11390642, ECO:0000269|PubMed:12446774, ECO:0000269|PubMed:12588868, ECO:0000269|PubMed:12660173, ECO:0000269|PubMed:12676583, ECO:0000269|PubMed:12676925, ECO:0000269|PubMed:12676962, ECO:0000269|PubMed:14657349, ECO:0000269|PubMed:14681223, ECO:0000269|PubMed:14988723, ECO:0000269|PubMed:15650047, ECO:0000269|PubMed:15665856, ECO:0000269|PubMed:15707391, ECO:0000269|PubMed:15870257, ECO:0000269|PubMed:19716789,

Main Category	Category	Terms
		ECO:0000269|PubMed:25083873, ECO:0000269|PubMed:31316063}.
General Annotations	UP_COMMENT_PTM	Proteolytically cleaved at the C-terminus by SPRTN during normal DNA replication, thereby promoting CHEK1 removal from chromatin and activating the protein kinase activity. {ECO:0000269|PubMed:31316063}.
General Annotations	UP_COMMENT_PTM	Ubiquitinated. Mono or diubiquitination promotes nuclear exclusion (By similarity). The activated form (phosphorylated on Ser-345) is polyubiquitinated at Lys-436 by some SCF-type E3 ubiquitin ligase complex containing FBXO6 promoting its degradation. Ubiquitination and degradation are required to terminate the checkpoint and ensure that activated CHEK1 does not accumulate as cells progress through S phase, when replication forks encounter transient impediments during normal DNA replication. 'Lys-63'-mediated ubiquitination by TRAF4 at Lys-132 activates cell cycle arrest and activation of DNA repair (PubMed:32357935). {ECO:0000250, ECO:0000269|PubMed:10859164, ECO:0000269|PubMed:11390642, ECO:0000269|PubMed:12446774, ECO:0000269|PubMed:12676583, ECO:0000269|PubMed:12676925, ECO:0000269|PubMed:12676962, ECO:0000269|PubMed:14681223, ECO:0000269|PubMed:14988723, ECO:0000269|PubMed:15650047, ECO:0000269|PubMed:15707391, ECO:0000269|PubMed:15870257, ECO:0000269|PubMed:19716789}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the protein kinase superfamily. CAMK Ser/Thr protein kinase family. NIM1 subfamily. {ECO:0000256|ARBA:ARBA00010791}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the protein kinase superfamily. CAMK Ser/Thr protein kinase family. NIM1 subfamily. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000269|PubMed:11836499, ECO:0000269|PubMed:12676962, ECO:0000269|PubMed:15311285, ECO:0000269|PubMed:15710331, ECO:0000269|PubMed:9278511}. Chromosome {ECO:0000269|PubMed:12676962, ECO:0000269|PubMed:31316063, ECO:0000269|PubMed:9382850}. Cytoplasm {ECO:0000269|PubMed:12676962}. Cytoplasm, cytoskeleton, microtubule organizing center, centrosome {ECO:0000269|PubMed:15311285}. Note=Nuclear export is mediated at least in part by XPO1/CRM1 (PubMed:12676962). Also localizes to the centrosome specifically during interphase, where it may protect centrosomal CDC2 kinase from inappropriate activation by cytoplasmic CDC25B (PubMed:15311285). Proteolytic cleavage at the C-terminus by SPRTN promotes removal from chromatin (PubMed:31316063). {ECO:0000269|PubMed:12676962, ECO:0000269|PubMed:15311285, ECO:0000269|PubMed:31316063}.

Main Category	Category	Terms
General Annotations	UP_COMMENT_SUBUNIT	Interacts (phosphorylated by ATR) with RAD51 (PubMed:15665856). Interacts with and phosphorylates CLSPN, an adapter protein that regulates the ATR-dependent phosphorylation of CHEK1 (PubMed:16963448). Interacts with BRCA1 (PubMed:11836499). Interacts with and phosphorylates CDC25A, CDC25B and CDC25C (PubMed:9278511). Interacts with FBXO6, which regulates CHEK1 (PubMed:19716789). Interacts with PPM1D, which regulates CHEK1 through dephosphorylation (PubMed:15870257). Interacts with TIMELESS; DNA damage-dependent (PubMed:15798197). Interacts with FEM1B; activates CHEK1 in response to stress (PubMed:19330022). Interacts with TLK1 (PubMed:12660173). Interacts with XPO1 and YWHAZ (PubMed:12676962). Interacts with CDK5RAP3; antagonizes CHEK1 (PubMed:19223857). {ECO:0000269|PubMed:11836499, ECO:0000269|PubMed:12660173, ECO:0000269|PubMed:12676962, ECO:0000269|PubMed:12766152, ECO:0000269|PubMed:15665856, ECO:0000269|PubMed:15707391, ECO:0000269|PubMed:15798197, ECO:0000269|PubMed:15870257, ECO:0000269|PubMed:16963448, ECO:0000269|PubMed:19223857, ECO:0000269|PubMed:19330022, ECO:0000269|PubMed:19716789, ECO:0000269|PubMed:9278511}.
General Annotations	UP_COMMENT_SUBUNIT	Interacts (phosphorylated by ATR) with RAD51. Interacts with and phosphorylates CLSPN, an adapter protein that regulates the ATR- dependent phosphorylation of CHEK1. Interacts with BRCA1. Interacts with and phosphorylates CDC25A, CDC25B and CDC25C. Interacts with FBXO6, which regulates CHEK1. Interacts with PPM1D, which regulates CHEK1 through dephosphorylation. Interacts with TIMELESS; DNA damage- dependent. Interacts with FEM1B; activates CHEK1 in response to stress. Interacts with TLK1. Interacts with XPO1 and YWHAZ. Interacts with CDK5RAP3; antagonizes CHEK1. {ECO:0000256|ARBA:ARBA00064089}.
General Annotations	UP_COMMENT_SUBUNIT	[Isoform 1]: Isoform 1 associates with isoform 2, the interaction is disrupted upon phosphorylation by ATR. {ECO:0000269|PubMed:22184239}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Expressed ubiquitously with the most abundant expression in thymus, testis, small intestine and colon. {ECO:0000269|PubMed:9278511, ECO:0000269|PubMed:9382850}.
Identifiers	ENTREZ_GENE_ID	1111
Identifiers	ENSEMBL_GENE_ID	ENSG00000149554
Identifiers	UNIPROT_ID	Q6W8P8_HUMAN
Identifiers	UNIPROT_ID	E9PKQ3_HUMAN
Identifiers	UNIPROT_ID	E9PQW7_HUMAN
Identifiers	UNIPROT_ID	E9PJI4_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	E9PRU7_HUMAN
Identifiers	UNIPROT_ID	E9PPA5_HUMAN
Identifiers	UNIPROT_ID	E7EPP6_HUMAN
Identifiers	UNIPROT_ID	CHK1_HUMAN
Identifiers	UNIPROT_ID	B4DT73_HUMAN
Identifiers	UNIPROT_ID	A0A087WT52_HUMAN
Interactions	DRUGBANK	XL844
Interactions	DRUGBANK	Enzastaurin
Interactions	DRUGBANK	CHIR-124
Interactions	DRUGBANK	N-{5-[4-(4-METHYLPIPERAZIN-1-YL)PHENYL]-1H-PYRROLO[2,3-B]PYRIDIN-3-YL}NICOTINAMIDE
Interactions	DRUGBANK	3-(5-{[4-(AMINOMETHYL)PIPERIDIN-1-YL]METHYL}-1H-INDOL-2-YL)QUINOLIN-2(1H)-ONE
Interactions	DRUGBANK	2,2'-{[9-(HYDROXYIMINO)-9H-FLUORENE-2,7-DIYL]BIS(OXY)}DIACETIC ACID
Interactions	DRUGBANK	(2S)-1-AMINO-3-[(5-NITROQUINOLIN-8-YL)AMINO]PROPAN-2-OL
Interactions	DRUGBANK	2-(cyclohexylamino)benzoic acid
Interactions	DRUGBANK	3-(5-{[4-(AMINOMETHYL)PIPERIDIN-1-YL]METHYL}-1H-INDOL-2-YL)-1H-INDAZOLE-6-CARBONITRILE
Interactions	DRUGBANK	(3Z)-6-(4-HYDROXY-3-METHOXYPHENYL)-3-(1H-PYRROL-2-YLMETHYLENE)-1,3-DIHYDRO-2H-INDOL-2-ONE
Interactions	DRUGBANK	5-ETHYL-3-METHYL-1,5-DIHYDRO-4H-PYRAZOLO[4,3-C]QUINOLIN-4-ONE
Interactions	DRUGBANK	(5-{3-[5-(PIPERIDIN-1-YLMETHYL)-1H-INDOL-2-YL]-1H-INDAZOL-6-YL}-2H-1,2,3-TRIAZOL-4-YL)METHANOL
Interactions	DRUGBANK	1-(5-CHLORO-2-METHOXYPHENYL)-3-{6-[2-(DIMETHYLAMINO)-1-METHYLETHOXY]PYRAZIN-2-YL}UREA
Interactions	DRUGBANK	(3-ENDO)-8-METHYL-8-AZABICYCLO[3.2.1]OCT-3-YL 1H-PYRROLO[2,3-B]PYRIDINE-3-CARBOXYLATE
Interactions	DRUGBANK	18-CHLORO-11,12,13,14-TETRAHYDRO-1H,10H-8,4-(AZENO)-9,15,1,3,6-BENZODIOXATRIAZACYCLOHEPTADECIN-2-ONE
Interactions	DRUGBANK	1-(5-CHLORO-2,4-DIMETHOXYPHENYL)-3-(5-CYANOPYRAZIN-2-YL)UREA
Interactions	DRUGBANK	4-(6-{[(4-METHYLCYCLOHEXYL)AMINO]METHYL}-1,4-DIHYDROINDENO[1,2-C]PYRAZOL-3-YL)BENZOIC ACID
Interactions	DRUGBANK	4-[3-(1H-BENZIMIDAZOL-2-YL)-1H-INDAZOL-6-YL]-2-

Main Category	Category	Terms
		METHOXYPHENOL
Interactions	DRUGBANK	(2R)-1-[(5,6-DIPHENYL-7H-PYRROLO[2,3-D]PYRIMIDIN-4-YL)AMINO]PROPAN-2-OL
Interactions	DRUGBANK	(2R)-3-{[(4Z)-5,6-DIPHENYL-6,7-DIHYDRO-4H-PYRROLO[2,3-D]PYRIMIDIN-4-YLIDENE]AMINO} PROPANE-1,2-DIOL
Interactions	DRUGBANK	N-(5,6-DIPHENYLFURO[2,3-D]PYRIMIDIN-4-YL)GLYCINE
Interactions	DRUGBANK	(5,6-DIPHENYL-FURO[2,3-D]PYRIMIDIN-4-YLAMINO)-ACETIC
Interactions	DRUGBANK	3-AMINO-3-BENZYL-[4.3.0]BICYCLO-1,6-DIAZANONAN-2-ONE
Interactions	DRUGBANK	3-(1H-BENZIMIDAZOL-2-YL)-1H-INDAZOLE
Interactions	DRUGBANK	2-[5,6-BIS-(4-METHOXY-PHENYL)-FURO[2,3-D]PYRIMIDIN-4-YLAMINO]-ETHANOL
Interactions	DRUGBANK	2-[(5,6-DIPHENYLFURO[2,3-D]PYRIMIDIN-4-YL)AMINO]ETHANOL
Interactions	DRUGBANK	REL-(9R,12S)-9,10,11,12-TETRAHYDRO-9,12-EPOXY-1H-DIINDOLO[1,2,3-FG:3',2',1'-KL]PYRROLO[3,4-I][1,6]BENZODIAZOCINE-1,3(2H)-DIONE
Interactions	DRUGBANK	1-[(2S)-4-(5-phenyl-1H-pyrazolo[3,4-b]pyridin-4-yl)morpholin-2-yl]methanamine
Interactions	DRUGBANK	N-(4-OXO-5,6,7,8-TETRAHYDRO-4H-[1,3]THIAZOLO[5,4-C]AZEPIN-2-YL)ACETAMIDE
Interactions	DRUGBANK	5,6,7,8-TETRAHYDRO[1]BENZOTHIENO[2,3-D]PYRIMIDIN-4(3H)-ONE
Interactions	DRUGBANK	[4-amino-2-(tert-butylamino)-1,3-thiazol-5-yl](phenyl)methanone
Interactions	DRUGBANK	2-(methylsulfanyl)-5-(thiophen-2-ylmethyl)-1H-imidazol-4-ol
Interactions	DRUGBANK	6-MORPHOLIN-4-YL-9H-PURINE
Interactions	DRUGBANK	1-[(2S)-4-(5-BROMO-1H-PYRAZOLO[3,4-B]PYRIDIN-4-YL)MORPHOLIN-2-YL]METHANAMINE
Interactions	DRUGBANK	Prexasertib
Interactions	DRUGBANK	Fostamatinib
Interactions	DRUGBANK	LY-2608204
Interactions	DRUGBANK	PF-477736
Interactions	PUBCHEM	Azd7762
Interactions	PUBCHEM	Bms-690514
Interactions	PUBCHEM	Kw-2449

Main Category	Category	Terms
Interactions	PUBCHEM	Rabusertib
Interactions	PUBCHEM	Lestaurtinib
Interactions	PUBCHEM	At-9283
Interactions	PUBCHEM	Dovitinib
Interactions	PUBCHEM	Nintedanib
Interactions	PUBCHEM	2-amino-2-cyclohexyl-N-[2-(1-methylpyrazol-4-yl)-9-oxo-3,10,11-triazatricyclo[6.4.1.04,13]trideca-1,4,6,8(13),11-pentaen-6-yl]acetamide
Interactions	PUBCHEM	Tak-901
Interactions	PUBCHEM	Silmitasertib
Interactions	PUBCHEM	PF-3758309
Interactions	PUBCHEM	K252a
Interactions	PUBCHEM	Ucn-01
Interactions	PUBCHEM	Refametinib
Interactions	PUBCHEM	Sch-900776
Interactions	PUBCHEM	Acetamide, N-[2-[6-[[4-(cyclobutylamino)-5-(trifluoromethyl)-2-pyrimidinyl]amino]-1,2,3,4-tetrahydronaphthalen-1,4-imin-9-yl]-2-oxoethyl]-
Interactions	PUBCHEM	Sunitinib
Interactions	PUBCHEM	Cerdulatinib Hydrochloride
Interactions	UP_KW_LIGAND	ATP-binding
Interactions	UP_KW_LIGAND	Nucleotide-binding
Literature	GENERIF_SUMMARY	utilization of peptide library analyses to develop specific, highly preferred substrate motifs for hCds1/Chk2 and Chk1
Literature	GENERIF_SUMMARY	radiation induced phosphotylation of chK1 is associated with p53-dependent cell cycle arrest pathways in tumor
Literature	GENERIF_SUMMARY	An ATM-independent S-phase checkpoint response involves CHK1 pathway.
Literature	GENERIF_SUMMARY	role in mediating G(2)M checkpoint in relation to the cellular resistance to the novel topoisomerase I poison BNP1350
Literature	GENERIF_SUMMARY	findings indicate that Chk1 directly phosphorylates Cdc25A during an unperturbed cell cycle, and that phosphorylation of Cdc25A by Chk1 is required for cells to delay cell cycle progression in response to double-strand DNA breaks
Literature	GENERIF_SUMMARY	Data suggest that the UVC-induced S checkpoint

Main Category	Category	Terms
		response of inhibition of replicon initiation is mediated by ATR signaling through Chk-1 and is independent of ATM, Nbs1, and Mre11.
Literature	GENERIF_SUMMARY	Chk1 is phosphorylated by ATM and NBS1 and is essential for ionizing radiation-induced inhibition of DNA synthesis and the G(2)/M checkpoint
Literature	GENERIF_SUMMARY	targets tousled like kinasese via DNA damage checkpoint
Literature	GENERIF_SUMMARY	CHK1 regulates the S phase checkpoint by coupling the physiological turnover and ionizing radiation-induced accelerated proteolysis of Cdc25A.
Literature	GENERIF_SUMMARY	Chk1 mediates both S and G2 checkpoint and is thus a more ubiquitous cell cycle checkpoint mediator than previously thought
Literature	GENERIF_SUMMARY	Chk1 protein complexes bind to single-stranded DNA and DNA ends and has a role in the repair of double strand breaks
Literature	GENERIF_SUMMARY	the G2 checkpoint in irradiated human cells derives from an overactivation of the ATR/CHK1 pathway.
Literature	GENERIF_SUMMARY	Chk1 expression is dispensable for somatic cell death and critical for sustaining G2 DNA damage checkpoint
Literature	GENERIF_SUMMARY	we show that Chk1 is essential for the suppression of TLK activity after replication block, but that ATR, Chk2 and BRCA1 are dispensable for TLK suppression.
Literature	GENERIF_SUMMARY	hyperoxia activates the ATR-Chk1 pathway and phosphorylates p53 at multiple sites in an ATM-independent manner, which is different from other forms of oxidative stress such as H2O2 or UV light.
Literature	GENERIF_SUMMARY	Data report the isolation and characterization of the 5' flanking region of the Chk1 gene, and show that this region has promoter activity.
Literature	GENERIF_SUMMARY	Data suggest that the Chk1/Cdc25A/14-3-3 pathway functions to prevent cells from entering into mitosis prior to replicating their genomes to ensure the fidelity of the cell division process.
Literature	GENERIF_SUMMARY	Chk1 is a novel Hsp90 client with a role in sensitization of tumor cells to replication stress
Literature	GENERIF_SUMMARY	data reveal the very different mode of regulation between CHK1 and CHK2
Literature	GENERIF_SUMMARY	CHEK1 interacts with XIAP protein during mitosis.
Literature	GENERIF_SUMMARY	The transient slow-down of DNA synthesis was abolished in cells lacking ATR, whereas CHK1-siRNA-treated cells, NBS1 or Fanconi anemia cells showed partial S-phase arrest.
Literature	GENERIF_SUMMARY	Brca1 and Chk1 are regulated by MCPH1 during DNA damage in tumor cell lines

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Formation of a stable complex between protein kinase CK2beta and Chk1 is not affected by the modification of Thr213 but it does require the presence of an active Chk1 kinase.
Literature	GENERIF_SUMMARY	the Chk1 pathway involves claspin, which functions in checkpoint control [review]
Literature	GENERIF_SUMMARY	results suggest that centrosome-associated Chk1 shields centrosomal Cdk1 from unscheduled activation by cytoplasmic Cdc25B, thereby contributing to proper timing of the initial steps of cell division, including mitotic spindle formation.
Literature	GENERIF_SUMMARY	These data further emphasise the role of Chk1 as a molecular target to inhibit in tumors with a defect in the G(1) checkpoint with the aim of increasing the selectivity and specificity of anticancer drug treatments.
Literature	GENERIF_SUMMARY	CHK1 inactivation in non-hodgkin's lymphoma occurs by loss of protein expression in a subset of aggressive variants alternatively to ATM, CHK2, and p53 alterations
Literature	GENERIF_SUMMARY	role of Chk1 in maintaining tumor cell viability following activation of the replication checkpoint
Literature	GENERIF_SUMMARY	Chk1 and Chk2 control the induction of the p53 related transcription factor p73 in response to DNA damage.
Literature	GENERIF_SUMMARY	Chk1 and Chk2 have roles in mismatch repair-dependent G2 arrest
Literature	GENERIF_SUMMARY	results indicate that the different radiosensitivities & G(2)checkpoint responses are independent of NHEJ, suggesting that CHK1-regulated checkpoint facilitates homologous recombination repair & protects cells from ionizing radiation-induced killing
Literature	GENERIF_SUMMARY	possible interplay between tumor protein p53 C-terminal phosphorylation and acetylation, and they provide an additional mechanism for the control of the activity of p53 by Checkpoint kinase 1 and Checkpoint kinase 2
Literature	GENERIF_SUMMARY	Chk1 downregulation can not only potentiate DNA-damaging agents, but also enhance the toxicity of anti-microtubule agents.
Literature	GENERIF_SUMMARY	Results show that the ATR/checkpoint kinase 1 pathway plays a predominant role in the response to topoisomerase I inhibitors in carcinoma cells.
Literature	GENERIF_SUMMARY	Expression of Chk1 blunted induction of gammaH2AX foci by Cdk inhibitors, indicating that Chk1 down-regulation was necessary to elicit the full phenotype.
Literature	GENERIF_SUMMARY	loss of PTEN and subsequent activation of AKT impair CHK1 through phosphorylation, ubiquitination, and reduced nuclear localization to promote genomic instability in tumor cells
Literature	GENERIF_SUMMARY	ATR phosphorylates Chk1 in cells expressing only a

Main Category	Category	Terms
		mutant of ATRIP that does not bind to RPA-ssDNA.
Literature	GENERIF_SUMMARY	Chk1 is required during normal S phase to avoid aberrantly increased initiation of DNA replication, thereby protecting against DNA breakage.
Literature	GENERIF_SUMMARY	results of the present study suggest existence of a checkpoint kinase 1-dependent mechanism for diallyl trisulfide-induced mitotic arrest in human prostate cancer cells
Literature	GENERIF_SUMMARY	Synthetized inhibitors are synthetically tractable and inhibit Chk1 by competing for its ATP site.
Literature	GENERIF_SUMMARY	cleavage of Claspin by caspase-7 inactivates the Chk1 signaling pathway
Literature	GENERIF_SUMMARY	Chk1 deficiency resulted in a premature onset of mitosis because of abnormal activation of cyclin B-Cdc2 and led to the activation of caspases 3 and 9 triggered by cytoplasmic release of cytochrome c.
Literature	GENERIF_SUMMARY	Data suggest that the microcephaly observed in patients with MCPH1 deficiencies is due to disruption of the ATR-BRCA1-Chk1 signaling pathway that is also disrupted in Seckel syndrome patients.
Literature	GENERIF_SUMMARY	Chk1 plays a primary role in the protection of cells from death induced by replication fork stress, whereas p21 mediates through its role in regulating entry into S phase
Literature	GENERIF_SUMMARY	Chk1 is post-translationally chaperoned to an active kinase. This reaction minimally requires Hsp90, Hsp70, Hsp40, Cdc37, and the protein kinase CK2.
Literature	GENERIF_SUMMARY	Gene repair directed by oligonucleotides activates a pathway that prevents corrected cells from proliferating in cell culture through the activation of Chk1 and Chk2.
Literature	GENERIF_SUMMARY	Chk1 and Claspin proteins regulate each other and thus ensure the proper cell cycle progression and replication checkpoint control.
Literature	GENERIF_SUMMARY	CHK1 maintains global replication fork stability in response to DNA damage amd DNA replication inhibitors.
Literature	GENERIF_SUMMARY	Chk1-dependent replication checkpoint which slows S phase progression can protect tumour cells against the cytotoxic effects of 5FU.
Literature	GENERIF_SUMMARY	These observations imply the regulation of mitotic Chk1 function through Chk1 phosphorylation at novel sites by Cdk1.
Literature	GENERIF_SUMMARY	Chk1-mediated G(2)/M cell-cycle arrest by the histamine H1 antagonist, loratidine in a tumor cell line.
Literature	GENERIF_SUMMARY	Inhibition of Chk1 by an RNAi approach resulted in an increase in cell death in p53-null cells, showing that the Chk1-dependent G(2) checkpoint protected cells that

Main Category	Category	Terms
		lacked a functional p53 pathway from oxidative stress
Literature	GENERIF_SUMMARY	Critical role in regulating chromosome stability, and in particular, common fragile site stability.
Literature	GENERIF_SUMMARY	the Chk1 pathway is regulated through both phosphorylation of Claspin and its controlled degradation.
Literature	GENERIF_SUMMARY	Checkpoint kinase 1 (Chk1) is required for mitotic progression through negative regulation of polo-like kinase 1 (Plk1)
Literature	GENERIF_SUMMARY	These results suggest that while TopBP1 is a general regulator of ATR, Claspin operates downstream of TopBP1 to selectively regulate the Chk1-controlled branch of the genotoxic stress response.
Literature	GENERIF_SUMMARY	Data suggest that degradation of Claspin by SCFbetaTrCP restrains Chk1 activation.
Literature	GENERIF_SUMMARY	the Chk1-mediated S-phase checkpoint targets initiation factor Cdc45 via a Cdc25A/Cdk2-independent mechanism
Literature	GENERIF_SUMMARY	Either chk1 or plk1-specific antisense oligodeoxynucleotides enhanced DNA damaging agent-induced apoptosis.
Literature	GENERIF_SUMMARY	Chk1 can be activated in a claspin-dependent manner leading to cyclin B1 down-regulation and providing the cells of an additional mechanism to inhibit mitosis entry in Hela cells.
Literature	GENERIF_SUMMARY	Results show under normal cell cycle conditions and in the absence of DNA damage, CHK1 constitutively phosphorylates CDC25B during interphase and thus prevents the premature initiation of mitosis.
Literature	GENERIF_SUMMARY	ataxia-telangiectasia-related protein kinase-Chk1-protein phosphatase 2A regulatory circuit functions to keep Chk1 in a low-activity state during an unperturbed cell division cycle
Literature	GENERIF_SUMMARY	Plk1 has a role in mitotic arrest, which is released by Chk1
Literature	GENERIF_SUMMARY	Chk1 functions to coordinate mitotic events through regulation of Cdc25B
Literature	GENERIF_SUMMARY	Interfering with PCNA function by overexpression of p21 mutant, impaired in its interaction with Cdks but not with PCNA, leads to ATR-dependent Chk1 activation.
Literature	GENERIF_SUMMARY	These results suggest a mechanism through which Chk1 could protect against tumorigenesis through its role in spindle checkpoint signaling.
Literature	GENERIF_SUMMARY	Chk1-mediated phosphorylation of FANCE is required for a function independent of FANCD2 monoubiquitination.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the Chk1 framshift mutations might be involved in colorectal tumourigenesis through a defect in response to DNA damage in a subset of sporadic colorectal cancers and HNPCCs.
Literature	GENERIF_SUMMARY	Novel substrates for Chk1 and Chk2 were screened using substrate target motifs determined previously by an oriented peptide library approach.
Literature	GENERIF_SUMMARY	Inhibition of checkpoint kinase 1 (Chk1) by RNA-mediated interference or drug treatment suppressed DNA-damage-induced centrosome amplification.
Literature	GENERIF_SUMMARY	Chk1 regulates the density of active replication origins during the vertebrate S phase.
Literature	GENERIF_SUMMARY	Phosphorylation of p53 (Ser15), Chk1 (Ser345), and Chk2 (Thr68) was also observed, suggesting that H/R activates p53 through checkpoint signals.
Literature	GENERIF_SUMMARY	zoledronate induces DNA damage and S phase arrest, accompanied by activation of the ATM/Chk1/cdc25 pathway in a human osteosarcoma cell line
Literature	GENERIF_SUMMARY	Chk1 is E2F regulated and highly expressed in triple-negative estrogen receptor /progesterone receptor / HER-2 breast carcinomas
Literature	GENERIF_SUMMARY	ATR/Chk1 pathway is activated at an early time point after the loss of Geminin and contributes to checkpoint arrest
Literature	GENERIF_SUMMARY	Centrosomal Chk1 induces G2/M cell cycle arrest and may evoke centrosome amplification, the latter possibly providing a backup mechanism for elimination of cells with impaired DNA damage checkpoints operating earlier during the cell cycle
Literature	GENERIF_SUMMARY	Frequent mutations in CHEK1(5%) damage response axis in colon cancers.
Literature	GENERIF_SUMMARY	protein kinase CK2 is involved in cell cycle regulation and indicate the mechanism by which CDC25A turnover might be regulated by Chk1 in the absence of DNA damage.
Literature	GENERIF_SUMMARY	This review of new findings demonstrates that DNA replication fork velocity (fork progression) is coordinated with inter-origin distance and that it can be actively slowed down by Chk1-dependent mechanisms in response to DNA damage.
Literature	GENERIF_SUMMARY	Loss of CHK1 function impedes DNA damage-induced FANCD2 monoubiquitination but normalizes the abnormal G2 arrest in Fanconi anemia.
Literature	GENERIF_SUMMARY	Abrogation of G2 checkpoint by Geldanamycin may play a central role in sensitizing p53-negative tumor cells to DNA-damaging and decatenation-inhibiting agents.
Literature	GENERIF_SUMMARY	suggests the unique role of Chk1 in preventing abrogation of the G2 checkpoint in p53+/+ cells

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Chk1 inhibition activated p53-regulated transcripts including Puma/BBC3 in tetraploid but not in diploid tumor cells
Literature	GENERIF_SUMMARY	A side-by-side comparison of the Hsp90-dependent chaperoning of Chk1 to that of the progesterone receptor (PR) was performed to show that these distinct types of clients have different chaperoning requirements.
Literature	GENERIF_SUMMARY	Results suggest that apoptosis induced by replication inhibitors in Chk1-depleted cells is dependent upon the helicase cofactor Cdc45.
Literature	GENERIF_SUMMARY	CHK1 is a histone H3 threonine 11 kinase that requires DNA damage-induced transcriptional repression.
Literature	GENERIF_SUMMARY	CHK1 was at least partially responsible for the lithium arrested hepatocellular carcinoma cell SMMC-7721 at G2/M checkpoint
Literature	GENERIF_SUMMARY	Chk1 and Chk2 regulate the functional associations between hBRCA2 and Rad51 in response to DNA damage
Literature	GENERIF_SUMMARY	These results demonstrate that phosphorylation of Claspin within the Chk1-binding domain is catalysed by an ATR-dependent kinase distinct from Chk1.
Literature	GENERIF_SUMMARY	Hsp72-depleted cells show defect in phosphorylation and activation of the protein kinase Chk1 by genotoxic stresses, such as UVC irradiation or camptothecin.
Literature	GENERIF_SUMMARY	findings show that BCL6 can directly bind to a DNA consensus element in the CHEK1 promoter and repress its expression in normal and malignant B-cells
Literature	GENERIF_SUMMARY	Chk1 appears to help defend genomic integrity through effects on several other pathways, including Fanconi anemia proteins, the mitotic spindle, and transcription of cell cycle-related genes [review]
Literature	GENERIF_SUMMARY	p53 is activated by stimulation of mismatch repair in response to the misincorporation of deoxynucleotides into newly synthesized DNA, long before the lack of pyrimidine nucleoside triphosphates causes the rate of DNA synthesis to slow appreciably.
Literature	GENERIF_SUMMARY	a conserved proliferating cell nuclear antigen-interacting protein sequence in Chk1 is required for checkpoint function
Literature	GENERIF_SUMMARY	Data show that Chk1 and the Claspin-Timeless module of replication forks not only participate in ATR signaling, but also protect stressed forks independently of ATR.
Literature	GENERIF_SUMMARY	Chk1, but neither Chk2 nor MK2, is an important regulator of S phase arrest, and suggest that an additional kinase can contribute to the G(2) arrest.
Literature	GENERIF_SUMMARY	Chk1 is the primary signal transducer linking activation of the ATM/ATR kinases to Cdc25A destruction in

Main Category	Category	Terms
		response to ionizing radiation.
Literature	GENERIF_SUMMARY	Mcl-1, perhaps acting as an adaptor protein, in controlling the ATR-mediated regulation of Chk1 phosphorylation
Literature	GENERIF_SUMMARY	Chk1 is activated by caspase-mediated cleavage during apoptosis and might be implicated in enhancing apoptotic reactions rather than attenuating the ATR-Chk1 pathway
Literature	GENERIF_SUMMARY	These findings systematically dissect the differential roles of Chk1 and Chk2 in a favorable model pursuing camptothecin-driven DNA damage responses.
Literature	GENERIF_SUMMARY	the inhibition of the stress response sensor CHEK1 might contribute to the antineoplastic activity of specific DNA-damaging drugs
Literature	GENERIF_SUMMARY	study demonstrates that Chk1 inhibition results in the activating phosphorylation of p38 mitogen-activated protein kinase
Literature	GENERIF_SUMMARY	Cyclin B1 is an efficacy-predicting biomarker for antineoplastic Chk1 inhibitors.
Literature	GENERIF_SUMMARY	Chk1 protein is a molecular target to be inhibited in tumors with a defective G1 checkpoint to increase the selectivity of anticancer treatments.
Literature	GENERIF_SUMMARY	Chk1 is phosphorylated at Ser(317) by ATR resulting in stabilization of CKII, which in turn leads to phosphorylation of PTEN at Thr(383)
Literature	GENERIF_SUMMARY	HSP90 inhibition abrogates the topoisomerase I poison-induced G2/M checkpoint in p53-null tumor cells by depleting CHK1 and WEE1.
Literature	GENERIF_SUMMARY	Chk1-Ser286 and -Ser301 phosphorylation occurs not only in mitosis but also with stalled DNA replication and stimulation of the damage checkpoint.
Literature	GENERIF_SUMMARY	Chk1 affects Cdc25A via rapid phosphorylation and protein turnover, inhibition of Cdc25A transcription by p53-ATF3 is required for the maintenance of cell cycle arrest.
Literature	GENERIF_SUMMARY	Essential function of Chk1 can be uncoupled from DNA damage checkpoint and replication control
Literature	GENERIF_SUMMARY	ATR-Chk1 signalling pathway plays a major role in the regulation of death in response to DNA replication stress and that the Chk1-suppressed pathway protecting cells from replication stress is clearly distinguishable from that protecting cells from IR
Literature	GENERIF_SUMMARY	Results show that depletion of DNA polymerase alpha and not Polepsilon or Poldelta by siRNA induces phosphorylation of Chk1 on Ser345, thus phenocopying antimetabolite exposure.
Literature	GENERIF_SUMMARY	relevant target of caffeine is the ATR-Chk1 pathway and

Main Category	Category	Terms
		that inhibiting ATR or Chk1 might have promise in preventing or reversing UV damage
Literature	GENERIF_SUMMARY	Chk1 regulates origin firing predominantly by controlling Cdk2-Cdc25 function.
Literature	GENERIF_SUMMARY	Chk1 abundance is regulated by the Cul4A/DDB1 ubiquitin ligase during an unperturbed cell division cycle, in response to replicative stress and on heat shock protein 90 inhibition.
Literature	GENERIF_SUMMARY	Critical role of ATM/Chk1 in curcumin-mediated G2/M cell cycle arrest and apoptosis in pancreatic cancer cells.
Literature	GENERIF_SUMMARY	Suppression of the Chk1/Chk2 protein by shRNA constructs inhibited the growth of oral pulp cells, which indicates that these checkpoint proteins may be essential for normal cell growth.
Literature	GENERIF_SUMMARY	The structure, regulation, and functions of Chk1 and Chk2 in the DNA damage response, as well as their potential roles in human disease, is discussed.
Literature	GENERIF_SUMMARY	Cdh1 reciprocally regulates the Rb pathway through competing with E2F1 to bind the hypophosphorylated form of Rb.
Literature	GENERIF_SUMMARY	ablation of the protein kinase Chk1 rescues the p21 transcription elongation defect when cells are blocked in S phase
Literature	GENERIF_SUMMARY	Results show that a lack of microcephalin or pericentrin results in a loss of Chk1 from centrosomes with subsequently deregulated activation of centrosomal cyclin B-Cdk1.
Literature	GENERIF_SUMMARY	Data show that disruption of ATR and Chk1 in mismatch repair-deficient cells increased the number of chromosomal abnormalities and enhanced the sensitivity to treatment with 5-fluorouracil.
Literature	GENERIF_SUMMARY	Data show that Chk1 inhibitor PF-00477736 offers a therapeutic potential for the enhancement of taxane therapy.
Literature	GENERIF_SUMMARY	Down-regulating Chk1 and Chk2 may increase the apoptotic sensitivity to irradiation due to changes of the pattern of cell cycle specific apoptosis.
Literature	GENERIF_SUMMARY	XAF1 as a novel cell cycle regulator through modulating G(2)/M checkpoint and interaction with checkpoint kinase 1 in gastrointestinal cancer.
Literature	GENERIF_SUMMARY	Studied vertebrate cells with compromised Chk1 function; only one protein, the histone chaperone nucelophosmin, was altered consistently in line with changes in chromatin-associated Chk1 and increased in response to DNA damage.
Literature	GENERIF_SUMMARY	Fbx6-dependent Chk1 degradation contributes to S phase checkpoint termination and that a defect in this

Main Category	Category	Terms
		mechanism might increase tumor cell resistance to certain anticancer drugs.
Literature	GENERIF_SUMMARY	Cocrystal structures of the inhibitors bound to CHK1 reveal an allosteric site, unique to CHK1, located in the C-terminal domain and consisting of a shallow groove linked to a small hydrophobic pocket.
Literature	GENERIF_SUMMARY	U2OS cells lacking Chk1 undergo abnormal mitosis and fail to activate the spindle checkpoint, suggesting a role of Chk1 in this checkpoint.
Literature	GENERIF_SUMMARY	Data show that the ATR-dependent phosphorylation of Chk1, but not p53, is strongly stimulated by Claspin.
Literature	GENERIF_SUMMARY	Chk1 phosphorylation by Cdk1 participates in cytoplasmic sequestration of Chk1 activity, which releases Cdk1 inhibition in the nucleus and promotes mitotic entry
Literature	GENERIF_SUMMARY	Chk1 signalling causes centrosome amplification after ionizing radiation by upregulating Cdk2 activity through activating phosphorylation.
Literature	GENERIF_SUMMARY	antagonizing CHK1 kinase strongly reduces the clonogenic properties of high-DNA damage level acute myeloid leukemia samples
Literature	GENERIF_SUMMARY	phosphorylation of S331 of FANCD2 is mediated by CHK1
Literature	GENERIF_SUMMARY	Abrogation of G(2) arrest by inhibition of ATM and ATR, Chk1, and Wee1 suppressed JCV genome replication.
Literature	GENERIF_SUMMARY	Fhit is significantly directly correlated with absence of activated Chk1 in dysplasia, suggesting a connection between loss of Fhit and modulation of checkpoint activity.
Literature	GENERIF_SUMMARY	proliferating cell nuclear antigen, ATR, TopBP1, and Chk1 are recruited to chromatin in a MutLalpha- and MutSalpha-dependent fashion after N-methyl-N'-nitro-N-nitrosoguanidine treatment.
Literature	GENERIF_SUMMARY	gammaH2AX foci in Chk1-depleted cells may represent sites of persistent replication fork damage or abandonment that are unable to resume DNA synthesis but do not play a direct role in the Chk1 suppressed death pathway
Literature	GENERIF_SUMMARY	New insights into the role of Chk1 in the DNA damage response machinery could provide an opportunity for novel approaches to the development of Chk1 inhibitor strategies.
Literature	GENERIF_SUMMARY	NEK11 controls degradation of CDC25A by directly phosphorylating CDC25A on residues whose phosphorylation is required for beta-TrCP mediated CDC25A polyubiquitylation and degradation.
Literature	GENERIF_SUMMARY	In the absence of DNA polymerase eta, Chk1 kinase becomes essential for replication resumption by

Main Category	Category	Terms
		alternative pathways, via fork stabilization.
Literature	GENERIF_SUMMARY	This study suggests a possible role for Chk1 in the differentiation program of umbilical cord blood CD133+ hematopoietic stem cells.
Literature	GENERIF_SUMMARY	The DNA damage-activated kinases Chk1 and Chk2 may be involved in tau phosphorylation and toxicity in the pathogenesis of Alzheimer's disease.
Literature	GENERIF_SUMMARY	RPA-covered ssDNA not only supports recruitment and activation of ATR but also, through Tipin and Claspin, it plays an important role in the action of ATR on its critical downstream target Chk1
Literature	GENERIF_SUMMARY	Akt prevents the translocation of BRCA1 to DNA damage foci and, thereby, inhibits the activation of Chk1 following DNA damage
Literature	GENERIF_SUMMARY	ATM and Chk1/2-mediated phosphorylation of cdc25C plays a major role in cell cycle arrest induced by pectenotoxin 2.
Literature	GENERIF_SUMMARY	Genomic rearrangements of the CHK1 gene is associated with breast cancer.
Literature	GENERIF_SUMMARY	Inhibition of plk1 through dephosphorylation was dependent on AtM/Chk1 activity
Literature	GENERIF_SUMMARY	Data show that Chk1 activation as a key factor to mediate TK/GCV cytotoxicity.
Literature	GENERIF_SUMMARY	describe an in vitro checkpoint system in which replication protein A-single-stranded DNA and TopBP1 are essential for phosphorylation of Chk1 by the purified ATR-ATR-interacting protein complex.
Literature	GENERIF_SUMMARY	14-3-3gamma mediates the interaction between Chk1 and Cdc25A; this ternary complex formation has an essential function in Cdc25A phosphorylation and degradation to block premature mitotic entry after DNA damage.
Literature	GENERIF_SUMMARY	Data suggest that recovery from oxidative DNA damage-induced G(2) arrest requires Chk1.
Literature	GENERIF_SUMMARY	Chk1-mediated phosphorylation on BLM at Ser(646) might be a determinant for regulating subnuclear localization and could act as a marker for the activation status of BLM in response to DNA damage.
Literature	GENERIF_SUMMARY	Chk1 promotes replication fork progression during normal S phase by controlling replication origin activity
Literature	GENERIF_SUMMARY	ATM-Chk2 and ATR-Chk1 pathways have roles in DNA damage signaling and cancer [review]
Literature	GENERIF_SUMMARY	The positive rates of CHK1 were significantly higher in gastric cancer than that in different types of benign lesions.
Literature	GENERIF_SUMMARY	In response to ultraviolet radiation, Lats2 is

Main Category	Category	Terms
		phosphorylated by Chk1 and Chk2 at S408.
Literature	GENERIF_SUMMARY	our data suggest that inactivation of EI24 and CHEK1 through two independent mechanisms contributes to the development of CACX.
Literature	GENERIF_SUMMARY	Chk1 is expressed in human sperm, but differently in different semen quality groups; expression is correlated with sperm DNA damage and apoptosis.
Literature	GENERIF_SUMMARY	At low levels of replication stress, ATR/Chk1 suppresses origin initiation by inhibiting the activation of new replication factories (RF), thereby reducing the number of active RF, but allowing dormant origins to initiate within existing RF.
Literature	GENERIF_SUMMARY	analysis of a pathway through which Chk1 inhibition can mediate HDACi-induced mitotic entry and cell death
Literature	GENERIF_SUMMARY	TAp73 induction by nitric oxide
Literature	GENERIF_SUMMARY	studies have revealed a novel interplay between Chk1 kinase and Cdc14B phosphatase involving radiation-induced nucleolar shuttling to facilitate error-free cell cycle progression and prevent genomic instability
Literature	GENERIF_SUMMARY	Genomic DNA damage and ATR-Chk1 signaling determine oncolytic adenoviral efficacy in ovarian cancer cells.
Literature	GENERIF_SUMMARY	USP1-regulated and damage-specific DNA-binding protein 1 (DDB1)-dependent mechanisms that are involved in the downregulation of activated CHK1 in human cells.
Literature	GENERIF_SUMMARY	The data suggest that CHK1 deletion is an early genetic event in the development of breast cancer and can be detected at the ductal carcinoma in situ stage.
Literature	GENERIF_SUMMARY	Results indicate that glucose deprivation induces ubiquitin-mediated Chk1 degradation and defective checkpoint responses, implying its potential role in genomic instability and tumor development.
Literature	GENERIF_SUMMARY	Tethering DNA damage checkpoint mediator proteins topoisomerase IIbeta-binding protein 1 (TopBP1) and Claspin to DNA activates ataxia-telangiectasia mutated and RAD3-related (ATR) phosphorylation of checkpoint kinase 1 (Chk1).
Literature	GENERIF_SUMMARY	Data show that in MCF7 cells, both ATM (p53, p21) and ATR (Chk1) pathways are defective leading to a complete lack of G2 arrest in response to bleomycin and ICRF-193.
Literature	GENERIF_SUMMARY	a direct role of Chk1 in maintaining normal hematopoiesis in the bone marrow.
Literature	GENERIF_SUMMARY	These results suggest that human MutY homolog may interact with ATR and function as a mediator of Chk1 phosphorylation in response to DNA damage.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	nuclear but not centrosomal Chk1 contributes to correct timing of mitotic entry.
Literature	GENERIF_SUMMARY	identified a de novo role for ATR-Chk1 and DNA-PK signaling in B19V DNA replication in primary human erythroid progenitor cells
Literature	GENERIF_SUMMARY	Integrin subunits alpha5 and alpha6 regulate cell cycle by modulating the chk1 and Rb/E2F pathways to affect breast cancer metastasis.
Literature	GENERIF_SUMMARY	Data show that Mus81/Eme1-dependent DNA damage--rather than a global increase in replication-fork stalling--is the cause of incomplete replication in Chk1-deficient cells.
Literature	GENERIF_SUMMARY	These data indicate that Cdk-mediated phosphorytion is required for efficient Chk1 activation and multiple checkpoint proficiency.
Literature	GENERIF_SUMMARY	genetic disruption of PP5 is associated with enhanced and prolonged phosphorylation of a single serine (Ser-345) on Chk1
Literature	GENERIF_SUMMARY	Chk1 and FANCD2 signal equally and additively for senescence induction, while Chk1 is predominantly responsible for maintaining persistent cell cycle arrest.
Literature	GENERIF_SUMMARY	These compounds show a promising inhibitory profile on Chk1.
Literature	GENERIF_SUMMARY	Inhibition of checkpoint kinase 1 abrogates G2/M checkpoint activation and promotes apoptosis under heat stress.
Literature	GENERIF_SUMMARY	Data show that checkpoint kinase 1 (Chk1) is an important factor in the resistance of normal cells, and some transformed cells, to histone deacetylase inhibitors (HDACi)-induced cell death.
Literature	GENERIF_SUMMARY	The combination of Chk1 and PARP1 inhibition selectively radiosensitizes p53 mutant pancreatic cancer cells.
Literature	GENERIF_SUMMARY	WRN is required for the CHK1-mediated intra-S-phase checkpoint.
Literature	GENERIF_SUMMARY	Data indicate that SCH900776 is a selective Chk1 inhibitor that potently inhibits Chk1 and abrogates cell-cycle arrest.
Literature	GENERIF_SUMMARY	these data lend insight into the mechanism by which Chk1 enhances repair of DNA damage while at the same time repressing stalled replication fork restart.
Literature	GENERIF_SUMMARY	results suggest p90 RSK facilitates nuclear Chk1 accumulation through Chk1-Ser-280 phosphorylation and that this pathway plays an important role in the preparation for monitoring genetic stability during cell proliferation.
Literature	GENERIF_SUMMARY	The BRCA1/Chk1/p53 pathway mediated the

Main Category	Category	Terms
		upregulation of p21(Cip1/Waf1). We also demonstrated that the NO-mediated up-regulation of p21(Cip1/Waf1) was inversely correlated with the activation status of the p38MAPK pathway.
Literature	GENERIF_SUMMARY	Detailed characterization of this genetic system revealed an important role for Chk1 in maintaining a normal cell cycle, thereby preventing the growth of subclones with increased ploidy.
Literature	GENERIF_SUMMARY	Studies indicate that the ATR-Chk1-Cdc25 pathway is the sole checkpoint route in a majority of cancer cells.
Literature	GENERIF_SUMMARY	Results identify a crucial tumor suppressive role of miR-424 in the progression of cervical cancer at least partly via upreglating the expression of Chk1.
Literature	GENERIF_SUMMARY	These results suggest that Chk1 phosphorylates Mig-6 on Ser 251, resulting in the inhibition of Mig-6, and that Chk1 acts as a positive regulator of EGF signalling.
Literature	GENERIF_SUMMARY	Data show that the ability of checkpoint kinases Chk1 to promote replication fork progression on damaged DNA relies on its proliferating cell nuclear antigen (PCNA)-interacting motif, which is required for its release from chromatin after DNA damage.
Literature	GENERIF_SUMMARY	Human embryonic stem cells fail to activate CHK1 and commit to apoptosis in response to DNA replication stress.
Literature	GENERIF_SUMMARY	hypoxia can activate p53 through inactivation of MDMX by the ATR-Chk1-MDMX-14-3-3gamma pathway.
Literature	GENERIF_SUMMARY	CHK1 phosphorylation of SYK(L) promoted its subsequent proteasomal degradation.
Literature	GENERIF_SUMMARY	Data indicate that acute myeloid leukemia differentiation can be induced through ATP depletion and the subsequent activation of DNA damage signaling through an ATR/Chk1-dependent and p53-independent pathway.
Literature	GENERIF_SUMMARY	novel regulatory mechanisms that couple protein cellular localization with the checkpoint response and cell viability of Chk1.
Literature	GENERIF_SUMMARY	These data reveal that BRCA1 facilitates Chk1 phosphorylation and its partial chromatin dissociation following replication inhibition that is likely to be required for S-phase checkpoint signaling.
Literature	GENERIF_SUMMARY	Unique regulatory mechanisms of Chk1 phosphorylation; this suggests that expression of constitutively active Chk1 may represent a novel strategy to suppress tumor growth.
Literature	GENERIF_SUMMARY	findings identify claspin as an in vivo substrate for the BRCA1 E3 ligase and suggest that its modification selectively triggers CHK1 activation for the homology-directed repair of a subset of genotoxic lesions

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Loss-of-function and gain-of-function studies showed that miR-15 family members controlled the expression of WEE1 and CHK1.
Literature	GENERIF_SUMMARY	Findings suggest that checkpoint kinase 1 CHK1 mediates cell cycle checkpoint signals by both cis-autophosphorylation and trans-phosphorylation of downstream factors.
Literature	GENERIF_SUMMARY	CASP and non-CASP family proteases intricately regulate cleavage for up-regulation of CHK1 kinase activity during PCD.
Literature	GENERIF_SUMMARY	CRL4 is a major regulator of CHK1 stability. CRL4CDT2 targets CHK1 for ubiquitination in the nucleoplasm, and for PCNA-independent degradation. CHK1 is required for G2 arrest in CDT2-depleted cells.
Literature	GENERIF_SUMMARY	ATR and CHK1 promote chromosomal stability by different mechanisms
Literature	GENERIF_SUMMARY	MYC regulation of CHK1 and CHK2 promotes radioresistance in a stem cell-like population of nasopharyngeal carcinoma cells.
Literature	GENERIF_SUMMARY	Data propose that Chk1 and Mps1 jointly regulate Aurora-B, MCAK, Kif2b and Hec1 to correct merotelic attachments. These results suggest a role for Chk1 and Mps1 in error correction.
Literature	GENERIF_SUMMARY	IRAK1, CHEK1 and BUB1 may play an important role in ovarian cancer.
Literature	GENERIF_SUMMARY	UV-induced G2 checkpoint activation in this situation therefore relies on signalling via the p38 MAPK and ATR-Chk1 signalling cascades.
Literature	GENERIF_SUMMARY	A feedforward mechanism that leads to G2 arrest identifies ATR and Chk1 as molecular agents of the requisite checkpoint.
Literature	GENERIF_SUMMARY	Chk1 co-localizes and physically associates with Mad2 in cells both under unstressed conditions and after DNA damage.
Literature	GENERIF_SUMMARY	BTG3-dependent CHK1 ubiquitination contributes to its chromatin localization and activation and a defect in this regulation may increase genome instability and promote tumorigenesis
Literature	GENERIF_SUMMARY	these findings indicate that the novel Chk1 inhibitor MK-8776 markedly potentiates HDACI lethality in leukemia cells displaying various genetic backgrounds
Literature	GENERIF_SUMMARY	Through the ATR-Chk1-Cdc25C and Weel pathways were the central mechanisms for G2 /M phase arrest in human colon cancer cells.
Literature	GENERIF_SUMMARY	Chk1 pathway is activated in MCM7-overexpressed cells in a p53-dependent manner.
Literature	GENERIF_SUMMARY	Data indicate that 4-Hydroxynonenal (HNE) induces

Main Category	Category	Terms
		phosphorylation of telangiectasia mutated and Rad3-related protein (ATR) and checkpoint kinase 1 (Chk1).
Literature	GENERIF_SUMMARY	Pim kinases phosphorylate Chk1 and regulate its functions in acute myeloid leukemia.
Literature	GENERIF_SUMMARY	Che-1 depletion abolishes the ability of Chk1 to bind pericentrin and to localize at centrosomes, which, in its turn, deregulates the activation of centrosomal cyclin B-Cdk1 and advances entry into mitosis.
Literature	GENERIF_SUMMARY	DNA polymerase kappa-dependent DNA synthesis at stalled replication forks is important for CHK1 activation.
Literature	GENERIF_SUMMARY	common alleles of ATR and CHEK1 are not implicated in breast cancer risk or survival, but we cannot exclude effects of rare alleles and of common alleles with very small effect sizes.
Literature	GENERIF_SUMMARY	Abrogation of the G2 checkpoint through the inhibition of Chk1 may be used to selectively increase the cellular radiosensitivity of HPV-positive HNSCC without affecting the surrounding normal tissue.
Literature	GENERIF_SUMMARY	After replication stress, the checkpoint kinase Chk1 phosphorylates E2F6, leading to its dissociation from promoters. This promotes E2F-dependent transcription, which mediates cell survival by preventing DNA damage and cell death
Literature	GENERIF_SUMMARY	These findings indicate that DHM inhibits the growth of hepatocellular carcinoma (HCC) cells via G2/M phase cell cycle arrest through Chk1/Chk2/Cdc25C pathway
Literature	GENERIF_SUMMARY	Suppression of the CHK1 pathway by panobinostat is at least partially responsible for the synergistic antitumor interactions between panobinostat and the DNA damaging agents in high-risk neuroblastoma cells.
Literature	GENERIF_SUMMARY	elevated expression in lung adenocarcinomas correlates with poor patient survival
Literature	GENERIF_SUMMARY	It is a novel molecular targe of Wedelia chinensis extract in nasopharyngeal carcinoma cells.
Literature	GENERIF_SUMMARY	KIM-1 expression increased significantly in a manner that corresponded temporally and regionally with increased phosphorylation of checkpoint kinase 1 (Chk1) and STAT3 in acute kidney injury.
Literature	GENERIF_SUMMARY	CHK1-mediated phosphorylation of RASSF1A, at Serine 184, plays an important role in cell-cycle regulation and mutation of this CHK1 phosphorylation site in nasopharyngeal carcinoma has disease relevance.
Literature	GENERIF_SUMMARY	results establish a novel mechanism underlying the cooperative antileukemic activities of these drug combinations in which panobinostat suppresses expression of BRCA1, CHK1, and RAD51 to enhance cytarabine and daunorubicin sensitivities in AML cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Distal 11q loss is a useful biomarker in OSCC for radioresistance that can be reversed by ATR-CHEK1 pathway inhibition.
Literature	GENERIF_SUMMARY	miR-16 and miR-26a target checkpoint kinases Wee1 and Chk1 in response to p53 activation by genotoxic stress.
Literature	GENERIF_SUMMARY	It regulates signaling pathway in human lung cancercells.
Literature	GENERIF_SUMMARY	High CHEK1 expression is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Data indicate that inhibition of Src activity induces persistent activation of ataxia telangiectasia mutated (ATM)- and Rad3-related (ATR) and check point kinase 1 (Chk1).
Literature	GENERIF_SUMMARY	Chk1 gene significantly overexpressed in Triple Negative Breast Cancers. Depletion of Chk1 protein expression induced reduction of cell viability and led to mitotic cell death.
Literature	GENERIF_SUMMARY	STAT3 interrupts ATR-to-Chk1 signaling by promoting loss of Claspin, a protein that assists ATR to phosphorylate Chk1.
Literature	GENERIF_SUMMARY	Diallyl disulfide may specifically modulate Chk1 phosphorylation, and Diallyl disulfide induced G2/M phase arrest in BGC823 cells could result in part from Chk1-mediated inhibition of the Cdc25C/cyclin B1 pathway.
Literature	GENERIF_SUMMARY	Chk1 is required for metaphase-anaphase transition via regulating expression of Cdc20 and Mad2.
Literature	GENERIF_SUMMARY	Fos expression protects osteoblasts from RS and DNA damage likely through upregulation of Chk1 and facilitates transformation by Ras/E1A oncogenes.
Literature	GENERIF_SUMMARY	Combined inhibition of PDK1 and CHK1 represents a potentially effective therapeutic approach to reduce the growth of human glioblastoma.
Literature	GENERIF_SUMMARY	Data suggest carcinoma cells initiate S phase arrest/ autophagy as pro-survival responses to DNA damage (via antineoplastic drug); suppression of autophagy (via antineoplastic drug) enhances apoptosis via disruption of CHK1-dependent S phase arrest.
Literature	GENERIF_SUMMARY	Suggest a novel role for CHK1 in the control of Aurora B activation during DNA replication stress and support the idea that premature mitosis is a distinct cell fate triggered by the disruption of DNA replication when CHK1 function is suppressed.
Literature	GENERIF_SUMMARY	selective targeting of Chk1 and Chk2 by oncolytic adenovirus mutants was chosen to treat resistant tumor xenograft mice, and the maximum antitumoral efficacy was achieved with the combined co-abrogation of Chk1 and Chk2

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The results of this study provide the first clear information regarding the effects of Chk1 inhibition on radiation-induced G2 arrest, with special focus on the time dimension.
Literature	GENERIF_SUMMARY	These data support the further evaluation of Chk1 inhibitors in hematopoietic cancers as single agents as well as in combination with standard of care cytotoxic drugs.
Literature	GENERIF_SUMMARY	USP20 and Rad17 interact, and that this interaction is enhanced by UV exposure. We show that USP20 regulation of Rad17 is at the protein level in a proteasome-dependent manner. USP20 depletion results in poor activation of Chk1 protein by phosphorylation
Literature	GENERIF_SUMMARY	CCNB1 is activated by Chk1, exerts its oncogenic role in colorectal cancer cells, and may play a key role in the development of a novel therapeutic approach against colorectal cancer.
Literature	GENERIF_SUMMARY	Chk1 activity is required for BAK multimerization in association with PUMA during mitochondrial apoptosis
Literature	GENERIF_SUMMARY	Chk1 phosphorylates BLM-S502 to inhibit cullin-3-mediated BLM degradation during interphase.
Literature	GENERIF_SUMMARY	phosphorylation of Chk1 at ATR sites reduces the interaction between Chk1 and the MCM complex, facilitating chromatin release of phosphorylated Chk1, a critical step in the initiation and amplification of cell cycle checkpoint
Literature	GENERIF_SUMMARY	Our study provides compelling evidence that CHK1 plays a critical role in the anti-leukemic activity of MK-1775
Literature	GENERIF_SUMMARY	ZEB1 is an ATM substrate linking ATM to CHK1 and the mechanism underlying the association between epithelial-mesenchymal transition and radioresistance.
Literature	GENERIF_SUMMARY	Results show that CHK1 is overexpressed and critical for T-cell acute lymphoblastic leukemia proliferation and viability by downmodulating replication stress and preventing ATM/caspase-3-dependent cell death.
Literature	GENERIF_SUMMARY	Dual Tra2 protein depletion reduces expression of full-length CHK1 protein, results in the accumulation of the DNA damage marker gammaH2AX and decreased cell viability.
Literature	GENERIF_SUMMARY	Chk1 activation leads to phosphorylation of Smurf1 that enhances its self-degradation, hence resulting in a RhoB accumulation to promote apoptosis.
Literature	GENERIF_SUMMARY	we report that SPB07479 appear promising for further development of Chk1 inhibitors. This study also highlights the role of conserved water molecules in the active site of Chk1 for the successful identification of novel inhibitors

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	these compounds did not inhibit ATR catalytic activity in vitro, unlike typical ATP-competitive inhibitors, but acted in a mechanistically distinct manner to disable ATR-Chk1 function
Literature	GENERIF_SUMMARY	Findings suggest that the Chk1/Ribonucleotide reductase replication complex is a key component of the replication checkpoint.
Literature	GENERIF_SUMMARY	Report strong synergism observed by combining Chk1 and Wee1 inhibitors in preclinical models of mantle cell lymphoma.
Literature	GENERIF_SUMMARY	Inhibition of UNC45A expression reduced ChK1 activation and its tethering to the centrosome, events required for proper centrosome function.
Literature	GENERIF_SUMMARY	PhosphoChk1 and H2AX are useful biomarkers for ATR inhibition using a variety of immuno-detection methods, but timing may be critical
Literature	GENERIF_SUMMARY	these results suggest that breast cancer cells may rely on mTORC2-Chk1 pathway for survival
Literature	GENERIF_SUMMARY	ATR and CHEK1 are promising biomarkers and rational drug targets for personalized therapy in breast cancer.
Literature	GENERIF_SUMMARY	Results revealed a novel interplay between IKKepsilon, CHEK1 and p21 signaling in survival of ovarian cancer cells.
Literature	GENERIF_SUMMARY	Chk1 negatively regulates immunoglobulin somatic hypermutation and gene conversion.
Literature	GENERIF_SUMMARY	The suppression of PKC activates the Chk1-mediated mitotic exit checkpoint in Nf1-deficient cells, leading to the induction of apoptosis via mitotic catastrophe.
Literature	GENERIF_SUMMARY	Results show that Chk1 negatively regulates the Treslin-mediated loading of Cdc45 onto chromatin and thereby serves to antagonize the initiation of replication.
Literature	GENERIF_SUMMARY	Suppression of CHK1 by ETS Family Members Promotes DNA Damage Response Bypass and Tumorigenesis
Literature	GENERIF_SUMMARY	Data suggest that pCHK1 may have prognostic and predictive significance in breast cancer. Subcellular localisation of pCHK1 protein is related to its function.
Literature	GENERIF_SUMMARY	Findings suggest a role of checkpoint kinase 1 (CHK1) as an histone H3.3 serine 31 kinase.
Literature	GENERIF_SUMMARY	The findings implicate Aven in ATR-Chk1 signalling and point towards Chk1 inhibition as a strategy to sensitize human osteosarcomas to chemotherapy.
Literature	GENERIF_SUMMARY	The review summarizes Chk1 role as a multifaceted and versatile signaling factor that acts at ongoing forks and replication origins to determine the extent and quality of the cellular response to replication stress.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	normal DNA replication and replication checkpoint activation is regulated through the novel phosphorylation of MCM3 by Chk1
Literature	GENERIF_SUMMARY	CHEK1 was a direct target of miR-195, which decreased CHEK1 expression in lung cancer cells. High expression of CHEK1 in lung tumors was associated with poor overall survival.
Literature	GENERIF_SUMMARY	Our findings suggest that the addition of CHEK1 inhibitor increases the response of ovarian cancer cells to TPT. Furthermore, reduced dosages of both drugs achieved maximal cytotoxic effects by combining TPT with CHEK1 inhibitor.
Literature	GENERIF_SUMMARY	ATR activity in midbody-stage cells links postmitotic genome surveillance to abscission timing and that Chk1 integrates this and other signals upstream of Aurora B to regulate when the final step in the physical separation of daughter cells occurs.
Literature	GENERIF_SUMMARY	These findings indicated that mangiferin inhibits cell cycle progression through the ATR-Chk1 stress response DNA damage pathway, leading to cell cycle arrest at G2/M phase in leukemia cells
Literature	GENERIF_SUMMARY	nasopharyngeal carcinoma cells depend on CHK1 and WEE1 activity for growth
Literature	GENERIF_SUMMARY	mutations targeting the putative Chk1 KA1 domain confer constitutive biological activity by circumventing the need for ATR-mediated positive regulatory phosphorylation.
Literature	GENERIF_SUMMARY	Via activation of the ATM/ATR-Chk1/Chk2 check point pathway.
Literature	GENERIF_SUMMARY	Our study reveals a level of mechanistic complexity that offers conceptually important insights into the dynamic interplay between gemcitabine and CHK1 inhibition.
Literature	GENERIF_SUMMARY	DNA damage induces Chk1 phosphorylation on chromatin followed by releasing phospho-Chk1 from the chromatin into soluble nucleus and the cytoplasm where Chk1 activates the cell cycle checkpoints; and Chk1 is degraded and checkpoint signaling is terminated.
Literature	GENERIF_SUMMARY	ATR inhibition induces moderate ssDNA and triggers a DNA-PK and Chk1-mediated backup pathway to suppress origin firing.
Literature	GENERIF_SUMMARY	Isolate/characterize mantle cell lymphoma cell line resistance to Chk1 inhibitor PF-00477736.
Literature	GENERIF_SUMMARY	Chk1 is a predictive biomarker of radiotherapy resistance and early local recurrence.
Literature	GENERIF_SUMMARY	CHEK1 loss-of-function mutations have not been found in human tumors, and transgenic expression of Chek1 in mice promotes oncogene-induced transformation. [review]

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results support the inhibition of checkpoint kinase 1 (Chk1) as a new therapeutic strategy in acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	Inappropriate activation of CDK2 in S phase underlies the sensitivity of a subset of cell lines to Chk1 inhibitors.
Literature	GENERIF_SUMMARY	High CHK1 expression correlates with urinary bladder cancer.
Literature	GENERIF_SUMMARY	these results demonstrate a positive feedback loop involving Rad9A-dependend activation of Chk1.
Literature	GENERIF_SUMMARY	RAD9 has a prominent role in the ATR-Chk1 pathway that is necessary for successful formation of the damage-sensing complex and DNA damage checkpoint signaling.
Literature	GENERIF_SUMMARY	A new pathway of proliferation restriction for tetraploid untransformed cells that seems to be specific for loss of adhesion dependent cytokinesis failure involves Chk1 and p53 activation during G2.
Literature	GENERIF_SUMMARY	In conclusion, this study exemplifies cancer-specific synthetic lethality between two proteins in the same pathway and raises the prospect of combining ATR and CHK1 inhibitors as promising cancer therapy.
Literature	GENERIF_SUMMARY	Avoiding damage formation through invalidation of Mus81-Eme2 and Mre11, or preventing damage signaling by turning off the ATM pathway, suppresses the replication phenotypes of Chk1-deficient cells.
Literature	GENERIF_SUMMARY	Human induced pluripotent stem cells fail to activate CHK1 when exposed to DNA replication inhibitors and commit to apoptosis instead.
Literature	GENERIF_SUMMARY	Chk1's expression is controlled by p53 and RB/E2F1 at the transcriptional level.
Literature	GENERIF_SUMMARY	persistence of CHK1 levels in response to DNA damage in p53-deficient cancer cells, leads to CHK1-mediated activation of NF-kappaB and induction of NF-kappaB-regulated genes in cells and in associated tumor-derived microvesicles, both of which are abrogated by loss or inhibition of CHK1.
Literature	GENERIF_SUMMARY	We suggest that MNNG-stimulated ATR/CHK1 signaling stabilizes E2F3 by S124 phosphorylation, and then E2F3 together with NFY co-transactivate RRM2 expression for DNA repair.
Literature	GENERIF_SUMMARY	CHK1 overexpression is associated with T-cell and Hodgkin Lymphoma.
Literature	GENERIF_SUMMARY	This study shows that Chk1 indeed maintains a closed conformation in the absence of DNA damage through an intramolecular interaction between a region (residues 31-87) at the N-terminal kinase domain and the distal C terminus. A highly conserved Leu-449 at the C terminus is important for this intramolecular

Main Category	Category	Terms
		interaction.
Literature	GENERIF_SUMMARY	CHEK1-mediated DNA damage checkpoint has a role in the ESR2-NCF1-ROS pathway sensitization of esophageal cancer cells to 5-fluorouracil-induced cell death
Literature	GENERIF_SUMMARY	Results unveil a new aspect of PERK function and previously unknown roles for Claspin and Chk1 as negative regulators of DNA replication in the absence of genotoxic stress.
Literature	GENERIF_SUMMARY	These findings demonstrate an unsuspected requirement for a balanced nucleotide pool for optimal Chk1 activation both in unchallenged cells and in response to genotoxic stress.
Literature	GENERIF_SUMMARY	Data show that the checkpoint kinase 1/2 (Chk1/Chk2) inhibitor prexasertib (LY2606368) inhibits cell viability in B-/T-ALL cell lines.
Literature	GENERIF_SUMMARY	we demonstrate that CHK1 mRNA is overexpressed in two independent cohorts of medulloblastoma patient samples in comparison to normal cerebellum
Literature	GENERIF_SUMMARY	PM2.5 exposure strongly induced the activation of the ATR (ATR serine/threonine kinase)-CHEK1/CHK1 (checkpoint kinase 1) axis, which subsequently triggered TP53-dependent autophagy and VEGFA production in Beas-2B cells.
Literature	GENERIF_SUMMARY	Checkpoint kinase 1 and 2 signaling is important for apoptin regulation.
Literature	GENERIF_SUMMARY	High CHK1 expression is associated with increased radioresistance of non-small cell lung cancer.
Literature	GENERIF_SUMMARY	inhibition of Chk1 can potentiate activity of nucleoside analogs in TP53-mutated B-lymphoid cells
Literature	GENERIF_SUMMARY	monitoring CHEK1 expression could be used both as a predictor of outcome and as a marker to select AML patients for CHK1 inhibitor treatments.
Literature	GENERIF_SUMMARY	revealed more than 170 NFAT-associated proteins, half of which are involved in transcriptional regulation. Among them are many hitherto unknown interaction partners of NFATc1 and NFATc2 in T cells, such as Raptor, CHEK1, CREB1, RUNX1, SATB1, Ikaros, and Helios.
Literature	GENERIF_SUMMARY	PLAUR to be essential for activation of Checkpoint kinase 1 (CHK1); maintenance of cell cycle arrest after DNA damage in a TP53-dependent manner; expression, nuclear import and recruitment to DNA-damage foci of RAD51 recombinase, the principal protein involved in the homologous recombination repair pathway.
Literature	GENERIF_SUMMARY	ATR inhibition potentiated Chk1 inhibitor induced replication stress and cytotoxicity via the abrogation of ATR-dependent feedback activation of Chk1 induced by Chk1 inhibitor generated replication stress in tumor cell

Main Category	Category	Terms
		lines.
Literature	GENERIF_SUMMARY	Small molecule ATR and Chk1 inhibitors potently sensitize lymphoma cells to UVA radiation and induce a prominent apoptotic response
Literature	GENERIF_SUMMARY	HPV31 regulates RRM2 levels through expression of E7 and activation of the ATR-Chk1-E2F1 DNA damage response, which is essential to combat replication stress upon entry into S-phase.
Literature	GENERIF_SUMMARY	Chk1 inhibition with GDC-0425 in combination with gemcitabine was tolerated with manageable bone marrow suppression. The observed preliminary clinical activity warrants further investigation of this chemopotentiation strategy
Literature	GENERIF_SUMMARY	Genetic variants of the CHEK1 gene are significantly related to overall survival and disease-free survival of esophageal squamous cell carcinoma patients.
Literature	GENERIF_SUMMARY	disruption of IGF-1R signaling with small-molecule inhibitors or IGF-1 withdrawal partially abrogates both the phosphorylation and activation of CHK1 by ATR and the accompanying inhibition of chromosomal DNA synthesis in UVB-irradiated keratinocytes.
Literature	GENERIF_SUMMARY	Role of the CHK1-RAD51 signaling pathway in osteosarcoma cells.
Literature	GENERIF_SUMMARY	Date show that when Wee1 alone is inhibited, Chk1 suppresses CDC45 loading and thereby limits the extent of unscheduled replication initiation and subsequent S-phase DNA damage, despite very high CDK-activity.
Literature	GENERIF_SUMMARY	Histone deacetylases 1 and 2 cooperate in regulating BRCA1, CHK1, and RAD51 expression in acute myeloid leukemia cells.
Literature	GENERIF_SUMMARY	Suggest that activation of ATR/CHK1 signaling pathway is key for Epstein Barr virus-induced B-cell transformation.
Literature	GENERIF_SUMMARY	The findings reveal ATXN3 to be a novel deubiquitinase of Chk1, providing a new mechanism of Chk1 stabilization in genome integrity maintenance.
Literature	GENERIF_SUMMARY	AZD7762 demonstrates synergy with regard to inhibition of AR-CDC6-ATR-Chk1 signaling.
Literature	GENERIF_SUMMARY	these data demonstrate that prexasertib is a specific inhibitor of CHK1 in neuroblastoma and leads to DNA damage and cell death in preclinical models of this devastating pediatric malignancy.
Literature	GENERIF_SUMMARY	Rif1 can mediate MCM dephosphorylation at replication forks and that the stability of dephosphorylated replisomes strongly depends on Chk1 activity.
Literature	GENERIF_SUMMARY	Data show that protein phosphatase-1 alpha (PP1alpha) is required to maintain checkpoint kinase 1 (CHK1) in a dephosphorylated state and for the accelerated

Main Category	Category	Terms
		replication fork progression in Spi1/PU.1 transcription factor-overexpressing cells.
Literature	GENERIF_SUMMARY	Following DNA damage, addition of the TLK1 inhibitor, THD, or overexpression of NEK1-T141A mutant impaired ATR and Chk1 activation, indicating the existence of a TLK1>NEK1>ATR>Chk1 pathway. Indeed, overexpression of the NEK1-T141A mutant resulted in an altered cell cycle response after exposure of cells to oxidative stress, including bypass of G1 arrest and implementation of an intra S-phase checkpoint.
Literature	GENERIF_SUMMARY	These results indicate that, under stressful conditions, maintained mTORC1 signaling in cancer cells promotes survival by suppressing endogenous DNA damage, and may control cell fate through the regulation of CHK1.
Literature	GENERIF_SUMMARY	results show that HGF was involved in regulating Chk1 phosphorylation, and further demonstrate that AKT activity was responsible for this HGF-induced Chk1 phosphorylation.
Literature	GENERIF_SUMMARY	Chk1 was linked to DNA damage response bypass by suppressing JNK activation following oxidative stress, promoting cell cycle progression despite DNA damage.
Literature	GENERIF_SUMMARY	The ATR-Chk1 and ATM-Chk2 signalings in male breast cancer (MBC).
Literature	GENERIF_SUMMARY	expression of AURKA and CHEK1 was linked with detrimental outcome in patients. Our data describe a synthetic lethality interaction between CHEK1 and AURKA inhibitors with potential translation to the clinical setting
Literature	GENERIF_SUMMARY	This study reports the crystal structure of the human Chk1 putative kinase-associated 1 (KA1) domain, demonstrating striking structural homology with other sequentially diverse KA1 domains. Separately purified Chk1 kinase and KA1 domains are intimately associated in solution, which results in inhibition of Chk1 kinase activity.
Literature	GENERIF_SUMMARY	results suggest a Chk1-OGT-vimentin pathway that regulates the intermediate filament network during cytokinesis
Literature	GENERIF_SUMMARY	BRCA1 downregulation combined with CHK1 inhibition induced excessive amounts of DNA damage, resulting in an inability to complete the S-phase. Therefore, we suggest CHK1 inhibition as a strategy for targeting BRCA1- or CDK12-deficient tumors.
Literature	GENERIF_SUMMARY	Expression levels of phosphorylated cdc25A (p-cdc25A) and phosphorylated Chk1 (p-Chk1), belonging to the ATR pathway, were decreased by treatment with Dclk1 inhibitor LRRK2-IN-1 (LRRK), indicating Dclk1 involvement in the ATR pathway
Literature	GENERIF_SUMMARY	Cellular conditions, especially cell density, influenced the target engagement of V158411 for Chk1.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Xanthatin functions as a DNA-damaging agent in non-small cell lung carcinomas by activating Chk1-mediated DDR and lysosome-mediated degradation of Cdc25C.
Literature	GENERIF_SUMMARY	Study provides evidence that expression of CHEK1 protein is high in breast tumors arising in Nigerian women and is associated with aggressive cancer phenotypes and is a prognostic marker.
Literature	GENERIF_SUMMARY	our results identify a novel link between XRRA1 and the ATM/CHK1/2 pathway and suggest that XRRA1 is involved in a DNA damage response that drives radio- and chemoresistance by regulating the ATM/CHK1/2 pathway.
Literature	GENERIF_SUMMARY	Targeting the ATR/CHK1 axis could result in reversing of drug resistance in small cell lung cancer.
Literature	GENERIF_SUMMARY	Blocking apoptosis alone is insufficient to allow the subsequent outgrowth of primary B cells lacking CHK1 in vivo or B lymphoma lines in vitro, as these cells trigger p53- dependent cell cycle arrest in response to the accumulating DNA damage.
Literature	GENERIF_SUMMARY	The nuclear transcription factor Y subunit beta (NFYB)-E2F transcription factor 1 (E2F1) pathway displays a crucial role in the chemoresistance ofoxaliplatin-resistant colorectal cancer (OR-CRC) by inducing the expression and activation of checkpoint kinase 1 (CHK1), suggesting a possible therapeutic target for oxaliplatin resistance in CRC.
Literature	GENERIF_SUMMARY	CHEK1 inhibition targets breast cancer cells expressing higher levels of RNF126 by enhancing replication stress.
Literature	GENERIF_SUMMARY	Chk1 and 14-3-3 proteins cooperate to inactivate the transcriptional repressor functions of atypical E2F proteins. This mechanism might be of particular importance to cancer cells, since they are exposed frequently to DNA-damaging therapeutic reagents.
Literature	GENERIF_SUMMARY	DNA alkylation damage leads to ATR-Chk1 activation in cancer cells and ATR-Chk1 activation mitigates replication stress caused by mismatch repair-dependent processing of DNA damage.
Literature	GENERIF_SUMMARY	Chk1 and Chk2 are significantly expressed in human sperm. In case of sperm DNA damage, up-regulated Chk1 expression may enhance sperm apoptosis and lead to asthenospermia, while increased Chk2 expression may inhibit spermatogenesis and result in oligospermia.
Literature	GENERIF_SUMMARY	A pulse of gemcitabine and CHK1i followed by WEE1i durably suppressed tumor cell growth.
Literature	GENERIF_SUMMARY	CHK1 and CHK2 and their activated forms are frequently expressed in HGSC effusions, with higher expression following exposure to chemotherapy, and their expression is related to survival.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Chk1 phosphorylates Src-S51 to fully induce Src kinase activity and phosphorylated Src promotes formation of actin patches and stabilizes chromatin bridges.
Literature	GENERIF_SUMMARY	we conclude that cellular functions regulated by CHK1 are phosphorylation/localization-dependent and deregulation of CHK1 function occurs in VSCC and might contribute to tumorigenesis.
Literature	GENERIF_SUMMARY	control of CHK1 localization between the nucleus and cytoplasm.
Literature	GENERIF_SUMMARY	Studies indicate mechanism for translocation of Tuberous Sclerosis Complex (TSC complex) protein to lysosomes in response to DNA damage, which depends on ATM and Rad3-related protein (ATR)/checkpoint kinase 1 (Chk1)-mediated rhoB GTP-binding protein (RhoB) phosphorylation and sumoylation.
Literature	GENERIF_SUMMARY	CHK1 contains a beta-TrCP recognizable degron domain, which is phosphorylated by AMPK in response to glucose deprivation.
Literature	GENERIF_SUMMARY	these observations demonstrate that Chk1 T378/T382 auto-phosphorylation within the KA1 domain is linked to kinase activation and rapid proteasomal degradation, and suggest a non-canonical mechanism of regulation.
Literature	GENERIF_SUMMARY	Inhibition of checkpoint kinase 1 following gemcitabine-mediated S phase arrest results in CDC7- and CDK2-dependent replication catastrophe.
Literature	GENERIF_SUMMARY	These results indicate that, similar to its function in DNA damage response, Chk1 controls normal cell cycle progression mainly by inducing Cdc25A degradation.
Literature	GENERIF_SUMMARY	we report that WRN-deficient cells trigger an ATM signalling, which is responsible for CHK1 phosphorylation observed after prolonged Aph-induced replication perturbation
Literature	GENERIF_SUMMARY	Data implicate that CHK1 autophagic degradation is enhanced by HPV8, which may contribute to the oncogenic potential of the virus.
Literature	GENERIF_SUMMARY	we showed that PrKD1 phosphorylates all CDC25 isoforms (known substrates of CHEK1 kinases), independent from CHEK kinases, suggesting that direct phosphorylation of CDC25 by PrKD1 may be an alternate mechanism of G1 arrest.
Literature	GENERIF_SUMMARY	Fbxo6 confers drug-sensitization to cisplatin via inhibiting the activation of Chk1 in non-small cell lung cancer
Literature	GENERIF_SUMMARY	role with ATR in signaling mechanism that limits origin firing during unperturbed
Literature	GENERIF_SUMMARY	Data suggest that fork progression in the absence of Chk1 generates obstacles to the replisome, which negatively feedback on fork elongation contributing only mildly to the poor proliferation rates of Chk1-

Main Category	Category	Terms
		depleted cells.
Literature	GENERIF_SUMMARY	steady-state CHK1 activity safeguards its stability to maintain intrinsic checkpoint functions and ensure genome integrity and cell survival.
Literature	GENERIF_SUMMARY	CHK1 inhibition kills primary mouse and human hematopoietic stem cells via BCL2-regulated apoptosis.
Literature	GENERIF_SUMMARY	5-Aminoimidazole-4-carboxamide ribonucleoside - mediated cell differentiation was depended on the activation of the DNA damage-associated enzyme checkpoint kinase 1 (Chk1) induced by pyrimidine depletion.
Literature	GENERIF_SUMMARY	Cisplatin treatment resulted in increased autophosphorylation of ATR (T1989) and CHK1 (S345) phosphorylation that was markedly suppressed in the presence of withaferin A(WA). These results indicate that WA is an inhibitor of ATR in human breast cancer cells.
Literature	GENERIF_SUMMARY	Direct regulation of Chk1 protein stability by E3 ubiquitin ligase HUWE1.
Literature	GENERIF_SUMMARY	Differential properties of mitosis-associated events following CHK1 and WEE1 inhibitor treatments in human tongue carcinoma cells have been reported.
Literature	GENERIF_SUMMARY	we show that Cdc7 is required for Claspin-Chk1 interaction in human cancer cells by phosphorylating CKBD (Chk1-binding-domain) of Claspin.
Literature	GENERIF_SUMMARY	A functional variant in CHK1 contributes to increased risk of nasopharyngeal carcinoma in a Han Chinese population.
Literature	GENERIF_SUMMARY	Phosphatase POPX2 interferes with cell cycle by interacting with Chk1.
Literature	GENERIF_SUMMARY	Prevention of DNA Replication Stress by CHK1 Leads to Chemoresistance Despite a DNA Repair Defect in Homologous Recombination in Breast Cancer.
Literature	GENERIF_SUMMARY	The iron-sulfur helicase DDX11 promotes the generation of single-stranded DNA for CHK1 activation.
Literature	GENERIF_SUMMARY	Constitutive CHK1 Expression Drives a pSTAT3-CIP2A Circuit that Promotes Glioblastoma Cell Survival and Growth.
Literature	GENERIF_SUMMARY	Mitotic entry upon Topo II catalytic inhibition is controlled by Chk1 and Plk1.
Literature	GENERIF_SUMMARY	Expression and prognostic significance of the P53-related DNA damage repair proteins checkpoint kinase 1 (CHK1) and growth arrest and DNA-damage-inducible 45 alpha (GADD45A) in human oral squamous cell carcinoma.
Literature	GENERIF_SUMMARY	CHK1 Inhibition Is Synthetically Lethal with Loss of B-Family DNA Polymerase Function in Human Lung and

Main Category	Category	Terms
		Colorectal Cancer Cells.
Literature	GENERIF_SUMMARY	The Role of CHK1 Varies with the Status of Oestrogen-receptor and Progesterone-receptor in the Targeted Therapy for Breast Cancer.
Literature	GENERIF_SUMMARY	Ubiquitination of the DNA-damage checkpoint kinase CHK1 by TRAF4 is required for CHK1 activation.
Literature	GENERIF_SUMMARY	PIBF1 suppresses the ATR/CHK1 signaling pathway and promotes proliferation and motility of triple-negative breast cancer cells.
Literature	GENERIF_SUMMARY	Strong Correlation between the Expression of CHEK1 and Clinicopathological Features of Patients with Multiple Myeloma.
Literature	GENERIF_SUMMARY	Downregulation of CDC20 Increases Radiosensitivity through Mcl-1/p-Chk1-Mediated DNA Damage and Apoptosis in Tumor Cells.
Literature	GENERIF_SUMMARY	The Role of Chaperone-Mediated Autophagy in Cell Cycle Control and Its Implications in Cancer.
Literature	GENERIF_SUMMARY	High Expression of p21 as a Potential Therapeutic Target in Ovarian Clear-cell Carcinoma.
Literature	GENERIF_SUMMARY	Targeting TOPK sensitises tumour cells to radiation-induced damage by enhancing replication stress.
Literature	GENERIF_SUMMARY	CD138(-) multiple myeloma cells express high level of CHK1 which correlated to overall survival in MM patient.
Literature	GENERIF_SUMMARY	Chk1 promotes non-homologous end joining in G1 through direct phosphorylation of ASF1A.
Literature	GENERIF_SUMMARY	The mitotic checkpoint is a targetable vulnerability of carboplatin-resistant triple negative breast cancers.
Literature	GENERIF_SUMMARY	Overexpression of TopBP1, a canonical ATR/Chk1 activator, paradoxically hinders ATR/Chk1 activation in cancer.
Literature	GENERIF_SUMMARY	STAU2 protein level is controlled by caspases and the CHK1 pathway and regulates cell cycle progression in the non-transformed hTERT-RPE1 cells.
Literature	GENERIF_SUMMARY	PARP Activity Fine-tunes the DNA Replication Choreography of Chk1-depleted Cells.
Literature	GENERIF_SUMMARY	Structural basis for recruitment of the CHK1 DNA damage kinase by the CLASPIN scaffold protein.
Literature	GENERIF_SUMMARY	Critical role of SMG7 in activation of the ATR-CHK1 axis in response to genotoxic stress.
Literature	GENERIF_SUMMARY	[Cell Cycle Checkpoint Kinase and Drug Resistance of Lung Cancer].
Literature	GENERIF_SUMMARY	Maternal heterozygous mutation in CHEK1 leads to mitotic arrest in human zygotes.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Genetic variants of CHEK1, PRIM2 and CDK6 in the mitotic phase-related pathway are associated with nonsmall cell lung cancer survival.
Literature	GENERIF_SUMMARY	CHEK1 and circCHEK1_246aa evoke chromosomal instability and induce bone lesion formation in multiple myeloma.
Literature	GENERIF_SUMMARY	Genetic vulnerabilities upon inhibition of DNA damage response.
Literature	GENERIF_SUMMARY	Chk1 and the Host Cell DNA Damage Response as a Potential Antiviral Target in BK Polyomavirus Infection.
Literature	GENERIF_SUMMARY	Targeting CDC7 potentiates ATR-CHK1 signaling inhibition through induction of DNA replication stress in liver cancer.
Literature	GENERIF_SUMMARY	P90 ribosomal S6 kinase confers cancer cell survival by mediating checkpoint kinase 1 degradation in response to glucose stress.
Literature	GENERIF_SUMMARY	Chk1 Inhibition Potently Blocks STAT3 Tyrosine705 Phosphorylation, DNA-Binding Activity, and Activation of Downstream Targets in Human Multiple Myeloma Cells.
Literature	GENERIF_SUMMARY	CHEK1
Literature	GENERIF_SUMMARY	MiR-139-3p Targets CHEK1 Modulating DNA Repair and Cell Viability in Lung Squamous Carcinoma Cells.
Literature	GENERIF_SUMMARY	When more is less
Literature	GENERIF_SUMMARY	Chk1 Inhibition Ameliorates Alzheimer's Disease Pathogenesis and Cognitive Dysfunction Through CIP2A/PP2A Signaling.
Literature	GENERIF_SUMMARY	Aberrant Activation of Cell-Cycle-Related Kinases and the Potential Therapeutic Impact of PLK1 or CHEK1 Inhibition in Uterine Leiomyosarcoma.
Literature	GENERIF_SUMMARY	The SKP2-p27 axis defines susceptibility to cell death upon CHK1 inhibition.
Literature	GENERIF_SUMMARY	Thiosemicarbazones Can Act Synergistically with Anthracyclines to Downregulate CHEK1 Expression and Induce DNA Damage in Cell Lines Derived from Pediatric Solid Tumors.
Literature	GENERIF_SUMMARY	Identification of hub genes and their correlation with infiltration of immune cells in MYCN positive neuroblastoma based on WGCNA and LASSO algorithm.
Literature	GENERIF_SUMMARY	PKR-Mediated Phosphorylation of eIF2a and CHK1 Is Associated with Doxorubicin-Mediated Apoptosis in HCC1143 Triple-Negative Breast Cancer Cells.
Literature	GENERIF_SUMMARY	Claspin-Dependent and -Independent Chk1 Activation by a Panel of Biological Stresses.
Literature	GENERIF_SUMMARY	The Prevalence of CHEK1 and CHEK2 Mutations in

Main Category	Category	Terms
		Prostate Cancer
Literature	GENERIF_SUMMARY	Combined Aurora Kinase A and CHK1 Inhibition Enhances Radiosensitivity of Triple-Negative Breast Cancer Through Induction of Apoptosis and Mitotic Catastrophe Associated With Excessive DNA Damage.
Literature	GENERIF_SUMMARY	Bioinformatic analysis of CHEK1 as a marker of glomerular epithelial cell injury in diabetic nephropathy.
Literature	GENERIF_SUMMARY	Long Noncoding RNA ZBED5-AS1 Facilitates Tumor Progression and Metastasis in Lung Adenocarcinoma via ZNF146/ATR/Chk1 Axis.
Literature	GENERIF_SUMMARY	Mild replication stress causes premature centriole disengagement via a sub-critical Plk1 activity under the control of ATR-Chk1.
Literature	GENERIF_SUMMARY	CaMKK2 and CHK1 phosphorylate human STN1 in response to replication stress to protect stalled forks from aberrant resection.
Literature	GENERIF_SUMMARY	In silico and structure-based evaluation of deleterious mutations identified in human Chk1, Chk2, and Wee1 protein kinase.
Literature	GENERIF_SUMMARY	OTUB1 Targets CHK1 for Deubiquitination and Stabilization to Facilitate Lung Cancer Progression and Radioresistance.
Literature	GENERIF_SUMMARY	Spatial organization and functions of Chk1 activation by TopBP1 biomolecular condensates.
Literature	GENERIF_SUMMARY	The luciferase-based in vivo protein-protein interaction assay revealed that CHK1 promotes PP2A and PME-1 interaction.
Literature	GENERIF_SUMMARY	Biochemical studies suggest the functional interactions of this protein with cell cycle regulators. Cdc25A, Cdc25B, and Cdc25C can be phosphoryalted by this protein.
Pathways	KEGG_PATHWAY	Cell cycle
Pathways	KEGG_PATHWAY	p53 signaling pathway
Pathways	KEGG_PATHWAY	Cellular senescence
Pathways	KEGG_PATHWAY	Human T-cell leukemia virus 1 infection
Pathways	KEGG_PATHWAY	Human immunodeficiency virus 1 infection
Pathways	KEGG_PATHWAY	Viral carcinogenesis
Pathways	REACTOME_PATHWAY	Signaling by SCF-KIT
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Activation of ATR in response to replication stress
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Transcriptional Regulation by TP53
Pathways	REACTOME_PATHWAY	Regulation of TP53 Activity
Pathways	REACTOME_PATHWAY	HDR through Homologous Recombination (HRR)
Pathways	REACTOME_PATHWAY	DNA Double-Strand Break Repair
Pathways	REACTOME_PATHWAY	Homology Directed Repair
Pathways	REACTOME_PATHWAY	HDR through Homologous Recombination (HRR) or Single Strand Annealing (SSA)
Pathways	REACTOME_PATHWAY	Homologous DNA Pairing and Strand Exchange
Pathways	REACTOME_PATHWAY	Processing of DNA double-strand break ends
Pathways	REACTOME_PATHWAY	Presynaptic phase of homologous DNA pairing and strand exchange
Pathways	REACTOME_PATHWAY	TP53 Regulates Transcription of DNA Repair Genes
Pathways	REACTOME_PATHWAY	Regulation of TP53 Activity through Phosphorylation
Pathways	REACTOME_PATHWAY	G2/M DNA damage checkpoint
Pathways	REACTOME_PATHWAY	G2/M Checkpoints
Pathways	REACTOME_PATHWAY	Ubiquitin-Mediated Degradation of Phosphorylated Cdc25A
Pathways	REACTOME_PATHWAY	p53-Independent G1/S DNA Damage Checkpoint
Pathways	REACTOME_PATHWAY	G1/S DNA Damage Checkpoints
Pathways	REACTOME_PATHWAY	Cell Cycle Checkpoints
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	DNA Repair
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	Chk1/Chk2(Cds1) mediated inactivation of Cyclin B:Cdk1 complex
Pathways	REACTOME_PATHWAY	Transcriptional Regulation by E2F6
Pathways	REACTOME_PATHWAY	Signaling by Receptor Tyrosine Kinases
Pathways	WIKIPATHWAYS	miRNA regulation of DNA damage response
Pathways	WIKIPATHWAYS	Cell cycle
Pathways	WIKIPATHWAYS	Integrated cancer pathway
Pathways	WIKIPATHWAYS	Integrated breast cancer pathway
Pathways	WIKIPATHWAYS	Androgen receptor network in prostate cancer
Pathways	WIKIPATHWAYS	Retinoblastoma gene in cancer
Pathways	WIKIPATHWAYS	ATM signaling

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Pathways affected in adenoid cystic carcinoma
Pathways	WIKIPATHWAYS	ATR signaling
Pathways	WIKIPATHWAYS	ATM signaling in development and disease
Pathways	WIKIPATHWAYS	DNA IR double strand breaks and cellular response via ATM
Pathways	WIKIPATHWAYS	DNA IR damage and cellular response via ATR
Pathways	WIKIPATHWAYS	DNA repair pathways full network
Pathways	WIKIPATHWAYS	TROP2 regulatory signaling
Pathways	WIKIPATHWAYS	Molecular pathway for oxidative stress
Pathways	WIKIPATHWAYS	22q11 2 distal CNV
Pathways	WIKIPATHWAYS	DNA damage response
Protein Domains	INTERPRO	Prot_kinase_dom
Protein Domains	INTERPRO	Ser/Thr_kinase_AS
Protein Domains	INTERPRO	Kinase-like_dom_sf
Protein Domains	INTERPRO	Protein_kinase_ATP_BS
Protein Domains	INTERPRO	Chk1_catalytic_dom
Protein Domains	SMART	S_TKc